

Deep reinforcement learning via structure-aware cross-layer node representation for influence maximization in multilayer heterogeneous networks

Chenshuo Li^a, Jianxin Tang^{a,*}, Lijun Liu^a, Ping Wang^b, Binggang Ren^a, Bingkun Liang^a

^a School of Computer Science and Artificial Intelligence, Lanzhou University of Technology, Lanzhou, 730050, China

^b School of Traffic Transportation, Lanzhou Jiaotong University, Lanzhou, 730070, China

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ABSTRACT

Identifying a set of vital nodes to maximize global information spread in multilayer heterogeneous networks remains a critical yet challenging problem in influence maximization. Existing methods often overlook the structural characteristics where each layer contains distinct node sets, and fail to account for the complex many-to-many links between different layers. To address such challenges, a multilayer heterogeneous independent cascade model based on connectivity density is designed to capture the cross-layer information diffusion. Furthermore, this paper proposes a novel algorithm that integrates structure-aware cross-layer node representation into deep reinforcement learning (SACRL) to identify layer-specific optimal seed sets. Specifically, a structure-aware node encoder is designed to aggregate intra-layer and inter-layer topological features by combining a graph transformer with a cross-layer message passing mechanism, while a Dueling Deep Q-Network is employed to optimize the seed selection policy. Experimental evaluations on six real-world multilayer heterogeneous networks validate the effectiveness of the proposed SACRL and the performance improvement over the competitive state-of-the-art baselines.

1. Introduction

Nowadays, information propagation in social systems exhibits distinct cross-platform and multi-dimensional characteristics. Users participate in different platforms based on diverse interaction needs, such as establishing professional connections on career networks like LinkedIn, and consuming entertainment content on short-video sharing platforms like TikTok or YouTube [1]. Due to differences in functional positioning across social platforms, the user sets of different platforms often only partially overlap, and inter-layer relationships have evolved into many-to-many mapping structures [2]. Such structural complexity arises from the widespread phenomenon of account matrices or cross-platform semantic associations, where a node in one layer can simultaneously affect multiple target nodes in another layer [3]. Given the complexity, these multilayer structures can be modeled as multilayer heterogeneous networks, characterized by non-identical node sets across layers and many-to-many inter-layer connections.

To further illustrate structural heterogeneity, consider a two-layer professional network scenario. One layer consists of institutional organizations, while the other comprises individual professionals. These two node sets are disjoint and represent different functional entities. Furthermore, the many-to-many inter-layer edges can represent institutional affiliations. An organization can connect to multiple distinct individuals through employment or membership ties. When a new industry initiative emerges, the first-layer institutions can transmit this information downward to the second-layer professionals via employment or other relationships. These professionals then spread it across their personal networks. Conversely, when professionals adopt emerging technologies, the adoption can flow upward through the same connections, prompting the first-layer institutions to adopt the technology, which then spreads among institutions.

However, the classical influence maximization (IM) problem, which is formulated for monolayer networks [4], is inherently inadequate for cross-platform information diffusion scenarios. Existing solutions to the

* Corresponding author.

Email addresses: licshuo@lut.edu.cn (C. Li), tangjx@lut.edu.cn (J. Tang), liulj@lut.edu.cn (L. Liu), wangp@lzjtu.edu.cn (P. Wang), renbg@lut.edu.cn (B. Ren), liangbk@lut.edu.cn (B. Liang).

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IM problem, ranging from greedy algorithms to heuristic and machine learning-based methods, have predominantly focused on monolayer networks [5–8]. These methods typically simplify the social environment into a single isolated graph, and assume a unified channel for information diffusion. While such methods provide fundamental theoretical insights, they inherently overlook the heterogeneous nature of modern social interactions [9]. By treating layers in isolation or aggregating them into a simplistic structure, monolayer methods cannot capture the cross-platform propagation dynamics of influence [10], which can result in suboptimal seed selection in real-world scenarios where users operate in an interconnected manner. Thus, it is imperative to study the IM problem on multilayer heterogeneous networks (MHIM), which aims to identify layer-specific seed sets that can leverage the inter-layer edges to maximize global information dissemination [11,12].

To capture the multi-dimensional interactions, research has evolved toward multilayer networks, yet the majority of current studies remain confined to multiplex networks [13]. The multiplex paradigm relies on an idealized assumption of strict one-to-one alignment, where the node set is identical across all layers [14]. However, this assumption is far from the reality of multilayer heterogeneous networks. In real-world networks, node sets only partially overlap or are disjoint, and inter-layer connections form dense, irregular many-to-many mappings rather than simple inter-layer edges between identical nodes. Such structural heterogeneity implies that the influence diffusion paths are not merely a synchronized spread across aligned nodes but involve a complex traversal through diverse inter-layer connections [15]. Therefore, the IM problem in multilayer heterogeneous networks cannot be solved by simply superimposing monolayer solutions or assuming node alignment.

Furthermore, the research on propagation modeling for the MHIM problem remains underexplored. Existing studies lack a unified propagation model that can quantify diffusion dynamics driven by many-to-many inter-layer edges [13]. Many propagation models rely on simplified assumptions, such as synchronous activation of corresponding nodes across layers or randomly assigned inter-layer transmission probabilities [16,17]. While such models are designed for multiplex networks with aligned nodes, they break down in multilayer heterogeneous networks where the influence spread is constrained by the topological structure of many-to-many inter-layer connections. Therefore, these assumptions fail to capture how inter-layer edges affect propagation dynamics in multilayer heterogeneous networks.

Such multilayer heterogeneous network complexity also poses significant obstacles to existing seed selection algorithm design. Firstly, most methods aim to identify a unified set of seed nodes across all layers [18]. The “one-size-fits-all” strategy fundamentally conflicts with the reality of multilayer heterogeneous networks, where node sets across layers differ or only partially overlap. As a result, unified seed selection often leads to resource mismatch, since a node that has substantial influence in one layer may be marginal or even nonexistent in others, thereby failing to meet the distinct diffusion requirements of individual layers. Secondly, traditional heuristic methods and network embedding techniques typically struggle to capture the global topological information that arises from disjoint node sets and dense cross-layer bridges [19,20]. Consequently, such methods tend to produce seed sets that are locally optimal within individual layers, rather than fully exploiting the global influence propagation potential of the multilayer network.

In light of the characteristics of multilayer heterogeneous networks and the limitations of existing solutions, two key challenges arise in the MHIM problem: (1) How to design a refined propagation mechanism that models the varying diffusion probabilities across different layers based on inter-layer connection structures? (2) How to develop an effective seed selection algorithm that aggregates inter-layer information to identify optimal and layer-specific seed sets for each layer to maximize global influence spread?

To address the aforementioned challenges, a diffusion model for multilayer heterogeneous networks is introduced to capture both intra-layer and inter-layer propagation dynamics. Then, to solve the influence

maximization problem in multilayer heterogeneous networks, this paper proposes a novel algorithm termed SACRL, which integrates structure-aware cross-layer node representation into deep reinforcement learning. SACRL learns node representations through structure-aware cross-layer embeddings and optimizes the seed selection process using deep reinforcement learning (DRL). A Dueling Deep Q-Network (Dueling DQN) is trained to support decision-making for seed set selection. More specifically, the Dueling DQN utilizes a decoupled architecture to separately estimate state values and action advantages to achieve more stable Q-value estimation. Moreover, a graph transformer-based GNN encoder is utilized to aggregate feature information from intra-layer neighbors. Meanwhile, a cross-layer message passing mechanism is designed to aggregate global topological information across layers, which can enhance the expressiveness of the node embeddings. The key contributions of this paper can be summarized as follows:

- (1) A multilayer heterogeneous network independent cascade (MHIC) model based on inter-layer connectivity density is developed to characterize the many-to-many cross-platform information diffusion dynamics.
- (2) A specialized graph transformer-based GNN encoder is proposed. It integrates a cross-layer message passing mechanism and structure-aware feature augmentations to aggregate local structural patterns and global topological information.
- (3) The SACRL algorithm is proposed to solve the influence maximization problem on multilayer heterogeneous networks. The SACRL trains a Dueling DQN agent with structure-aware cross-layer embeddings to identify layer-specific optimal seed sets.

The paper is structured as follows. [Section 2](#) reviews prior work on influence maximization across monolayer, multiplex, and multilayer heterogeneous networks. [Section 3](#) provides the preliminaries, including problem formulation and essential definitions. [Section 4](#) details the proposed method. [Section 5](#) presents the experimental results. [Section 6](#) discusses the limitations of the method and future work, and [Section 7](#) concludes the paper.

2. Related work

2.1. Monolayer network influence maximization

The classical influence maximization problem is primarily concerned with identifying the optimal seed nodes in monolayer networks. Existing solutions can be classified into four types: greedy-based, heuristic-based, meta-heuristic-based, and machine learning-based methods [21,22]. Greedy-based methods leverage diffusion models like the independent cascade (IC) model to calculate the marginal gain of each node via thousands of Monte Carlo simulations. Leskovec et al. [23] introduced the cost-effective lazy forward (CELF) method, which exploits the submodularity property to eliminate redundant evaluations of marginal gains. While such greedy-based methods can guarantee effective results, the practical deployment on large-scale networks is often constrained by high computational costs.

To reduce computational overhead, many heuristic methods have been developed for IM. Teng et al. [24] proposed DGSIC, which constructs a dynamic communication prediction network based on 3D positional data and identifies critical nodes by integrating community structure, intra-community communication intensity, and inter-community communication scale. Although heuristic-based methods provide computational efficiency for influential node identification, such methods overlook the collaborative interactions among seed nodes, yielding suboptimal solutions. Moreover, meta-heuristic algorithms mimic natural evolutionary behaviors [22]. Kundu et al. [25] integrated a connectedness-based heuristic with learning automata-enabled ant colony optimization for group influence maximization. Although such

population-based optimization can balance exploration and exploitation, it remains susceptible to entrapment in local optima within limited iterations.

Furthermore, recent advances in machine learning have sparked growing interest in solving the IM problem [26]. Zhu et al. [27] proposed BiGDN with bidirectional neighborhood aggregation and multi-head attention for node representation, combined with Double Deep Q-network for seed selection. Gao et al. [28] proposed a cross-network node importance ranking framework. By introducing an influence-aware causal node embedding module and jointly optimizing the reconstruction and causal ranking losses, the framework achieves mutual enhancement between node representation learning and ranking objectives. Yang et al. [29] introduced MaDGNN, where a cascade-aware graph neural network learns node embeddings and cascade dynamics, while a Munchausen reinforcement learning module guides seed selection. Such machine learning-based methods exhibit strong generalization capabilities but require longer training times.

2.2. Multiplex network influence maximization

Monolayer networks have laid the foundation for influence maximization, but such networks inherently assume that individuals interact through only one type of relationship. In contrast, real-world social interactions are multifaceted, with individuals simultaneously connected through various channels such as friendship, professional collaboration, and online communication. To model multiple relationships, researchers have extended influence maximization to multiplex networks. Existing solutions on multiplex network influence maximization can be broadly divided into two categories based on seed selection strategies: unified seed selection and layer-specific seed selection. The former aims to identify a unified set of seed nodes across all layers, while the latter selects different seed sets for each layer independently.

Focusing on the unified seed selection methods, existing studies have proposed centrality-based metrics to quantify node importance in multiplex networks. Wang et al. [30] developed the CWIIF algorithm to identify influential nodes in multiplex networks by combining weighted intra-layer with inter-layer influence factors. Furthermore, Jiang et al. [31] proposed the DIRCI method to determine node importance in multi-layer networks by integrating dynamic influence range-based centrality, network layer centrality, and community-based centrality. Meanwhile, researchers have also focused on the application of gravity models in multiplex networks to refine node ranking. Zheng et al. [32] proposed the multiplex multi-attribute Laplacian gravity model that integrates multi-attribute Laplacian centrality with a gravity-based mechanism and a layer-wise weighting strategy to generate a global ranking of nodes. An et al. [33] introduced IFRSGC to estimate hidden relationship strengths in unweighted multilayer networks by fusing intrinsic and topological node features and then integrated the strengths into a weighted gravity centrality model for key node ranking. However, such centrality-based heuristics often fail to account for the overlap of influence spread among seeds, which can lead to suboptimal global spread.

Rather than relying on predefined centrality metrics, meta-heuristic algorithms have been employed to search for high-quality seed sets. Wang et al. [34] introduced a new weight-based multilayer influence approximation model, and then used a memetic algorithm to identify the optimal seed set in multiplex networks. Moreover, Wang et al. [17] developed a two-stage discrete particle swarm optimization framework for multiplex network influence maximization by utilizing the multilayer expected diffusion value as the fitness function. Similarly, Qu et al. [35] designed a memetic algorithm that combines global and local search to identify critical nodes in multiplex networks. Furthermore, Yang et al. [36] proposed an asynchronous distributed cooperative coevolutionary algorithm that employs multilayer community decomposition for problem decomposition and adaptive seed regulation for scalable influence maximization in multiplex networks. Such meta-heuristic algorithms

struggle to escape local optima induced by the complex structure of multiplex networks.

With the advancement of machine learning, novel methods have been introduced to capture complex structural patterns in multiplex networks. Zhang et al. [37] proposed a reverse random walk centrality method that employs the RMNE algorithm for multiplex network embedding and selects the nodes visited most frequently during the walk as seeds. By considering the characteristics of nodes and layers, Ju et al. [18] introduced a deep learning framework that leverages transformer-based node embeddings to predict node influence across the entire multiplex network structure. To address the functional dependency among inter-layer nodes, Gu et al. [38] introduced MultiDismantler to combine multiplex network representation with deep reinforcement learning to dismantle multiplex networks by identifying minimal node sets. Hassani et al. [39] presented GTBIM, which employs a GraphSAGE model for enriched node embeddings and a transformer encoder for adaptive layer weighting and influence estimation to solve the influence maximization problem in multiplex networks. Such methods can exhibit reduced generalizability when many-to-many edges exist between layers.

For layer-specific seed selection, Rao et al. [40] developed the CBIM to identify communities based on neighborhood similarity and select seed nodes from each layer via a quota-based approach. Compared to such community-aware methods, structural approaches leveraging tighter substructures have also been explored. Katukuri et al. [41] introduced CIM, a clique-based heuristic to select different seeds across layers through both inter-layer and intra-layer connections. Meanwhile, to address the issues of topological heterogeneity and optimal resource allocation, Tang et al. [42] proposed the LWCD method, which optimizes influence spread through layer-weighted budget allocation and community-based local dominance strategy in multiplex networks.

However, the assumptions of identical node sets and one-to-one inter-layer connections fail to account for nodes that exist only in certain layers, thereby causing the loss of layer-wise topological information. Furthermore, the neglect of many-to-many inter-layer connections restricts cross-layer diffusion paths to identical nodes. Consequently, the assumptions lead to biased estimations of global influence propagation and suboptimal seed node selection.

2.3. Multilayer heterogeneous network influence maximization

Multilayer heterogeneous networks relax the restrictive assumptions of node identity and simplified inter-layer connections prevalent in multiplex networks. By incorporating diverse node sets across layers and many-to-many inter-layer connections, the multilayer heterogeneous networks can provide a more realistic representation of complex social networks. Therefore, this has led to the study of influence maximization on multilayer heterogeneous networks.

To identify influential spreaders within such intricate networks, some studies have focused on developing centrality-based methods. Wan et al. [43] proposed the MLC method to identify influential nodes in multilayer heterogeneous networks by combining multirelational information with a hierarchical strategy that partitions nodes into core and auxiliary layers. Meanwhile, Li et al. [12] developed the coupled information feedback algorithm to measure the centrality of nodes in multilayer networks by introducing information feedback mechanisms to describe cross-layer interactions. Similarly, He et al. [44] introduced the NK-shell algorithm to determine key nodes in coupled transit networks by integrating global network structure with local features of first- and second-order neighboring nodes.

Furthermore, researchers have explored other approaches to solve the MHIM problem. Gao et al. [45] proposed a comprehensive identification framework for key nodes in multilayer heterogeneous aviation networks by aggregating multi-source structural and operational information through an improved TOPSIS-grey correlation analysis and

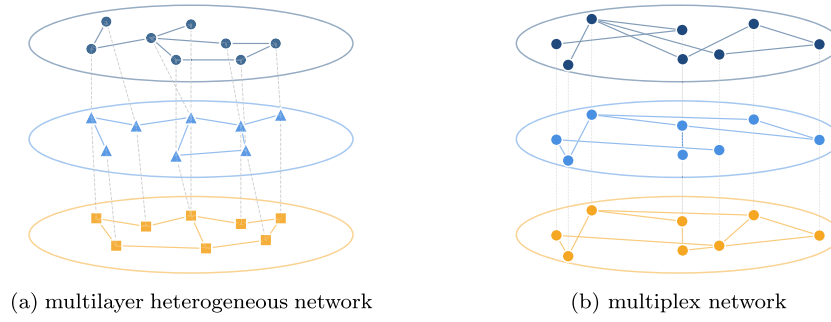


Fig. 1. Comparison of two network topologies.

gravity-based weighting. Li et al. [46] proposed a dynamic station-line centrality method for coupled transit networks, incorporating node attributes and community structure to capture time-varying node importance. Additionally, Chen et al. [47] developed a graph convolution-based evaluation framework to quantify node importance in multilayer heterogeneous networks by integrating cross-layer topological features with neighborhood aggregation mechanisms. Yang et al. [48] further introduced a graph representation learning approach to identify multilayer network hubs by capturing high-order node dependencies and cross-layer topological regularities via low-dimensional graph embeddings. To the best of our knowledge, identifying influential nodes in multilayer heterogeneous networks with many-to-many relationships remains under-explored, necessitating an effective framework.

3. Preliminary

3.1. Network topology

Multilayer networks provide a framework to characterize complex interactions among multiple systems. Depending on node alignment and connection patterns, these structures can be broadly categorized into multiplex networks, which assume identical node sets across all layers with one-to-one inter-layer alignments, and multilayer heterogeneous networks, which accommodate distinct node populations and many-to-many cross-layer interactions. An illustration comparing these two network topologies is provided in Fig. 1. Multilayer heterogeneous networks allow each layer to possess a unique user set that may only partially overlap or be entirely disjoint. Therefore, the structure captures scenarios like cross-platform social media, where a user on one platform may be connected to several different users on another.

3.2. Problem definition

Consider a multilayer heterogeneous network represented as a collection of graphs $\mathcal{G} = \{G_1, G_2, \dots, G_L\}$, where L denotes the total number of distinct layers, each corresponding to a specific social platform. Each layer is characterized by an intra-layer graph $G_i = (V_i, E_i)$, where V_i represents the set of users in layer i , and $E_i \subseteq V_i \times V_i$ captures the social interactions within the platform. The cross-platform interactions are encoded by the set of inter-layer edges $E^{inter} \subseteq \bigcup_{i \neq j} (V_i \times V_j)$, enabling cross-platform influence propagation across different users. The node and edge sets of the entire network are denoted by $V = \bigcup_{i=1}^L V_i$ and $E = \left(\bigcup_{i=1}^L E_i\right) \cup E^{inter}$, respectively.

In this study, the objective is to identify a collection of seed sets $S = \{S_1, S_2, \dots, S_L\}$, where each $S_i \subseteq V_i$ is selected from its corresponding layer under a predefined per-layer budget constraint $|S_i| = k$. When all selected seeds across layers are activated, information propagates through both intra-layer and inter-layer edges under the specific multilayer network diffusion model. Formally, the multilayer heterogeneous

influence maximization problem is defined by Eq. (1):

$$S^* = \arg \max_{\substack{S_i \subseteq V_i \\ |S_i| = k \\ i=1, \dots, L}} \sigma\left(\bigcup_{i=1}^L S_i\right) \quad (1)$$

where $\sigma(\cdot)$ denotes the influence spread function that evaluates the total number of activated nodes across all L layers at the termination of the diffusion process, and S_i represents the seed set selected from the i -th layer subject to the seed budget constraint $|S_i| = k$. The layer-specific budget ensures that each layer independently initiates a cascade that can subsequently propagate to other layers via inter-layer edges. In practice, giving each layer the same budget k is like putting the same number of influencers on every social platform in a real marketing campaign.

3.3. Propagation mechanism

The influence spread function $\sigma(\cdot)$ is grounded in the following propagation process. Each node $v \in V$ carries a binary activation state: either *inactive* ($x_v = 0$) or *active* ($x_v = 1$), where activation is irreversible. At $t = 0$, all seed nodes are set active and all remaining nodes are inactive. At each subsequent step, every newly activated node makes independent stochastic attempts to activate each of its currently inactive neighbors—both within the same layer and across layers via inter-layer edges—succeeding with layer-dependent probabilities. The process terminates when no new activation occurs, and $\sigma(S)$ denotes the expected total number of active nodes across all L layers at termination. The specific intra-layer and inter-layer activation probabilities are formally defined in Section 4.1.

4. The proposed algorithm

4.1. Multilayer heterogeneous network independent cascade model

Information diffusion in multilayer networks exhibits greater complexity than spreading dynamics in monolayer networks [15]. This intricacy stems from the heterogeneous topological interconnection structures across different platform layers. Users who are active in one layer may engage in cross-platform information exchange, which may set off information cascades across multiple social networks. Therefore, the interconnections between layers serve as a vital determinant in the diffusion process. The likelihood of inter-layer diffusion is correlated with the structural connectivity patterns between layers [49]. Densely interconnected layer pairs are likely to facilitate more frequent cross-platform information transfer, while sparsely connected layers exhibit weaker propagation tendencies. When implementing the IC model on each layer in isolation, it oversimplifies the diffusion processes in multilayer networks and fails to capture the intricate nature of inter-layer interactions [13]. Therefore, it is crucial to combine monolayer spreading dynamics with topology-aware inter-layer diffusion mechanisms to model the diffusion process.

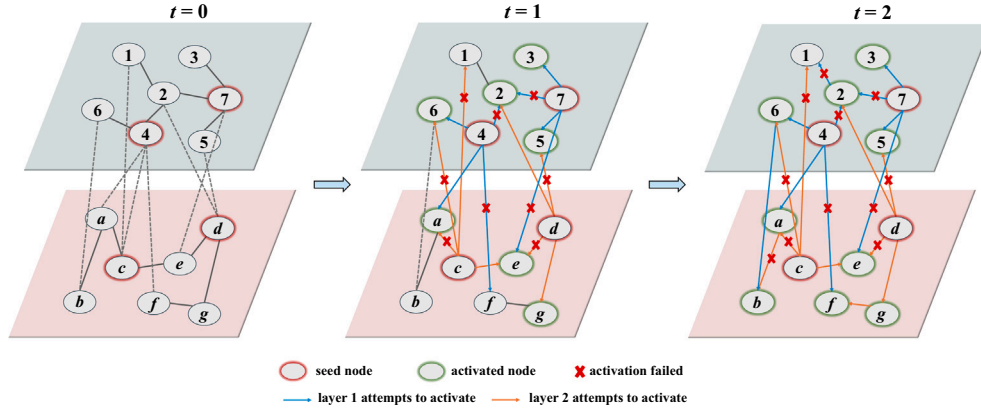


Fig. 2. An example of the MHIC model.

Given a multilayer heterogeneous network $\mathcal{G}$, the multilayer heterogeneous network independent cascade model is proposed to capture topology-driven cross-layer information propagation. The MHIC model considers both intra-layer and inter-layer propagation mechanisms. In intra-layer propagation, the IC model is adopted, where a node activated at time step t attempts to activate each of its unactivated neighbors within the same layer with probability p , and only successfully activated neighbors will participate in subsequent propagation at time step $t + 1$. In inter-layer propagation, once a node u is activated at time step t in layer i , it attempts to activate its connected nodes in other layers j ($j \neq i$) through existing inter-layer edges $(u, v) \in E_{ij}$ with probability q_{ij} at time step $t + 1$. The diffusion process terminates when no new nodes are activated in any layer, and the total number of activated nodes $\sigma(S)$ is then computed. The inter-layer propagation probability q_{ij} is defined in Eq. (2).

$$q_{ij} = \frac{|E_{ij}|}{|V_i| \times |V_j|} \quad (2)$$

where $|E_{ij}|$ denotes the number of inter-layer edges connecting layer i and layer j , and $|V_i|$ and $|V_j|$ represent the number of nodes in each respective layer. The inter-layer connectivity density q_{ij} quantifies the likelihood of cross-platform information spread and captures the heterogeneity of inter-layer coupling strengths, as real-world multilayer systems often exhibit vastly different connection densities across layer pairs. This density-based formulation is consistent with theoretical findings on spreading processes in multilayer networks [49], which identify macro-level inter-layer coupling strength as the primary driver of cross-layer cascade dynamics. Notably, q_{ij} varies across different layer pairs, directly reflecting the structural coupling strengths between distinct platform pairs.

Fig. 2 illustrates the information diffusion process under the MHIC model. At the initial time step $t = 0$, seed nodes $\{4, 7\}$ in layer 1 (gray) and seed nodes $\{c, d\}$ in layer 2 (pink) are activated. The dashed lines between the two layers represent inter-layer connections, which determine the inter-layer propagation probability q_{12} . At time step $t = 1$, the intra-layer propagation occurs: within layer 1, node 4 successfully activates nodes 6 and 7, while node 7 activates nodes 3 and 5, while neither node 4 nor node 7 activates node 2. Simultaneously, in layer 2, node c activates node e , and node d activates node g . The cross-layer propagation occurs: specifically, node 4 in layer 1 successfully activates node a in layer 2 via cross-layer propagation, while node d in layer 2 propagates its influence to node 2 in layer 1. All other cross-layer activation attempts fail. At time step $t = 2$, node 2 fails to activate node 1, while node g activates node f . Regarding cross-layer propagation, node 6 successfully activates node b in the other layer. This process demonstrates how the MHIC model captures the interaction between intra-layer propagation and topology-driven cross-layer cascading effects.

4.2. SACRL for the MHIM problem

To address the influence maximization problem in multilayer heterogeneous networks under the MHIC model, we introduce the SACRL algorithm that combines structure-aware node representation learning with deep reinforcement learning to identify layer-specific optimal seed sets. SACRL establishes a specialized structure-aware representation learning framework integrated with deep reinforcement learning. First, a multilayer heterogeneous network embedding module is designed. Rather than utilizing conventional attention mechanisms, this module features a graph transformer encoder augmented with a cross-layer message passing mechanism to process the dense, many-to-many inter-layer connection topologies. Additionally, it incorporates specific feature augmentations to ensure the resulting representations accurately capture both intra-layer topological importance and asymmetric inter-layer propagation potential. Second, a Dueling DQN module is employed to decompose the action-value function into separate state-value and action-advantage streams. This module utilizes an expected influence estimator as a proxy reward. SACRL identifies optimal layer-specific seed sets and maximizes the global influence spread across the multilayer heterogeneous network. The overall framework is illustrated in Fig. 3.

4.2.1. Multilayer heterogeneous network embedding module

For the IM task in multilayer heterogeneous networks, node embeddings must accurately reflect the structural importance and potential influence of nodes across multiple layers. To this end, a multilayer heterogeneous network embedding module is designed that integrates handcrafted structural features with a Graph Transformer encoder to generate node representations.

To capture topological properties, a five-dimensional structural feature vector $\mathbf{f}_v \in \mathbb{R}^5$ is computed for each node v in the multilayer heterogeneous network. Five complementary centrality measures are selected to characterize node importance from distinct structural perspectives. Degree centrality quantifies direct connectivity as an indicator of local influence capacity. Average neighbor degree reflects the propensity of a node to connect to well-connected neighbors, distinguishing nodes attached to influential hubs from those in peripheral regions. Eigenvector centrality captures recursive importance by accounting for the centrality of neighbors. PageRank models influence through a random-walk perspective. Finally, the K-shell index reveals the hierarchical core-periphery structure, identifying nodes in densely connected cores.

Formally, for a node $v \in V_i$ in layer $G_i = (V_i, E_i)$ with intra-layer neighbor set $\mathcal{N}_i(v)$, the local features based on the immediate neighborhood are defined by Eq. (3):

$$f_v^{(1)} = \frac{d_i(v)}{|V_i| - 1}, \quad f_v^{(2)} = \frac{1}{d_i(v)} \sum_{u \in \mathcal{N}_i(v)} d_i(u) \quad (3)$$

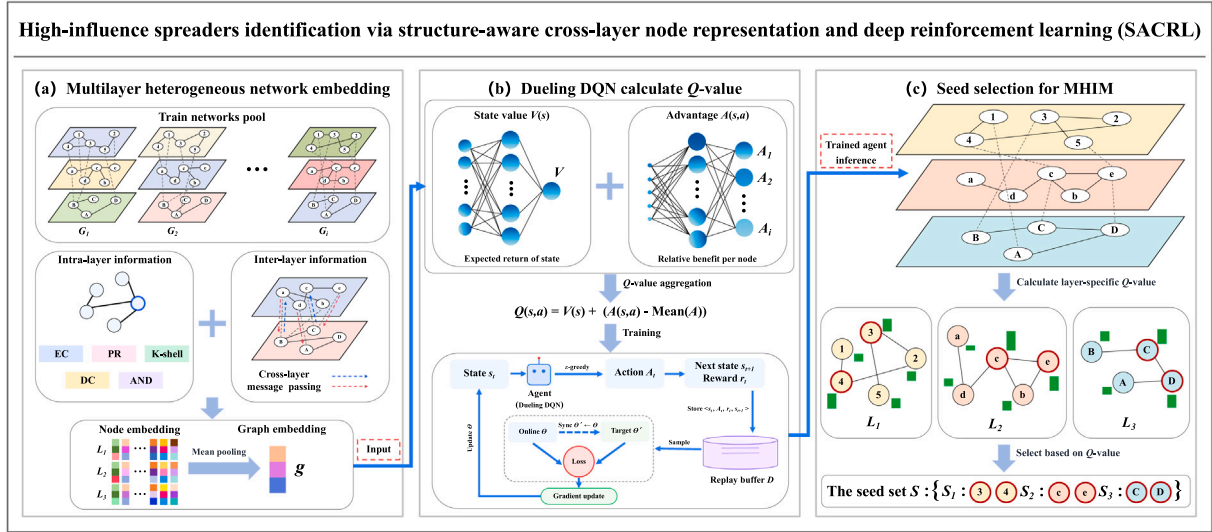


Fig. 3. The framework of the SACRL.

where $d_i(v)$ denotes the degree of node v in layer G_i . Next, the global features, which capture node importance from the perspective of the entire network topology, are calculated by Eq. (4):

$$\begin{aligned} f_v^{(3)} &= \frac{1}{\lambda_i} \sum_{u \in \mathcal{N}_i(v)} f_u^{(3)}, \\ f_v^{(4)} &= \alpha \sum_{u \in \mathcal{N}_i(v)} \frac{f_u^{(4)}}{d_i(u)} + \frac{1 - \alpha}{|V_i|}, \\ f_v^{(5)} &= k_s, \quad v \in \text{core}_i(k_s) \end{aligned} \quad (4)$$

where λ_i is the principal eigenvalue of the adjacency matrix for layer G_i , $\alpha = 0.85$ is the damping factor for PageRank, and $\text{core}_i(k_s)$ denotes the set of nodes belonging to the k_s -core structure of layer G_i . Unlike the degree feature, the features $f_v^{(3)}$ and $f_v^{(4)}$ are recursive in nature, characterizing node importance through the structural prestige of their neighbors. This global perspective, combined with the core-periphery depth revealed by the $f_v^{(5)}$, provides the initial topological profile that distinguishes nodes even in networks with locally homogeneous degree distributions.

Each feature dimension $j \in \{1, \dots, 5\}$ is normalized to the range $[0, 1]$ by Eq. (5). The layer-wise normalization ensures feature scales across networks of varying sizes and densities. The normalized features implicitly encode layer-level topological characteristics, which can enable the model to distinguish nodes from layers with different connectivity patterns. Beyond static structural features, two binary indicators expressed by Eq. (6) are incorporated to reflect whether a node has been selected as a seed and whether it is currently covered.

$$\tilde{f}_v^{(j)} = \frac{f_v^{(j)} - \min_{u \in V_i} f_u^{(j)}}{\max_{u \in V_i} f_u^{(j)} - \min_{u \in V_i} f_u^{(j)}} \quad (5)$$

$$f_v^{\text{cov}} = \mathbb{I}[v \in C], \quad f_v^{\text{seed}} = \mathbb{I}[v \in S_i^{(t)}] \quad (6)$$

where $S_i^{(t)} \subseteq V_i$ denotes the current seed set for layer G_i at step t , and C represents the covered set containing all nodes within the immediate neighborhood range of selected seeds. The two features are dynamically updated at each decision step, enabling the agent to perceive the current coverage status and avoid redundant selections.

Finally, each node is represented by a 70-dimensional feature vector $\mathbf{x}_v$ formed by concatenating three components: five structural metrics, two binary indicators, and a 63-dimensional vector of ones. The structural metrics capture complementary aspects of topological importance.

The two binary indicators are dynamically updated during decision-making to provide current state information. To alleviate the limitation of sparse 7-dimensional features, the input is augmented with a 63-dimensional constant vector. After the first linear projection, this vector acts as a learnable bias with 63 independent parameters, richer than a scalar bias. This shifts the ELU pre-activations of structurally peripheral nodes, whose centrality-based features are near zero, into the effective gradient regime, preventing low-gradient saturation during training. The feature design offers a sufficiently expressive input space for the graph transformer encoder to learn structural representations of nodes.

Given the initial node feature matrix $\mathbf{X} \in \mathbb{R}^{|V| \times d}$ where $d = 70$, a Graph Transformer encoder is employed to learn the node representations. The encoder operates on the multilayer heterogeneous network topology with edge set E , enabling information propagation across layers. For node v at the ℓ -th encoding level, the embedding $\mathbf{h}_v^{(\ell)}$ is updated via multi-head attention-weighted message passing. The intermediate levels ($\ell < N_L$) produce embeddings in $\mathbb{R}^{d_h \times H}$ where d_h denotes the hidden dimension per head and H is the number of attention heads, while the final level outputs embeddings in $\mathbb{R}^{d_h}$. In each attention head $m \in \{1, \dots, H\}$, the query, key, and value projections are computed by Eq. (7):

$$\mathbf{q}_{v,m}^{(\ell)} = \mathbf{W}_{Q,m}^{(\ell)} \mathbf{h}_v^{(\ell-1)}, \quad \mathbf{k}_{u,m}^{(\ell)} = \mathbf{W}_{K,m}^{(\ell)} \mathbf{h}_u^{(\ell-1)}, \quad \mathbf{v}_{u,m}^{(\ell)} = \mathbf{W}_{V,m}^{(\ell)} \mathbf{h}_u^{(\ell-1)} \quad (7)$$

where $\mathbf{W}_{Q,m}^{(\ell)}, \mathbf{W}_{K,m}^{(\ell)}, \mathbf{W}_{V,m}^{(\ell)} \in \mathbb{R}^{d_h \times d_{in}^{(\ell)}}$ are learnable projection matrices and $d_{in}^{(\ell)}$ is the input dimension of level ℓ .

The attention coefficient $\alpha_{vu,m}^{(\ell)}$ from neighbor u to node v in head m is computed by the softmax function which normalizes across both intra-layer and inter-layer neighborhoods:

$$\alpha_{vu,m}^{(\ell)} = \frac{\exp(\mathbf{q}_{v,m}^{(\ell)\top} \mathbf{k}_{u,m}^{(\ell)} / \sqrt{d_h})}{\sum_{w \in \mathcal{N}(v)} \exp(\mathbf{q}_{v,m}^{(\ell)\top} \mathbf{k}_{w,m}^{(\ell)} / \sqrt{d_h})} \quad (8)$$

where $\mathcal{N}(v) = \mathcal{N}_i(v) \cup \mathcal{N}_{inter}(v)$ denotes the neighbor set of node v , including both intra-layer and inter-layer neighbors. The inter-layer edges enable cross-layer information propagation, allowing the model to capture dependencies between layers in the multilayer heterogeneous network. Therefore, the attended output for each head aggregates information from both intra-layer and inter-layer neighbors, explicitly

modeling cross-layer message passing by Eq. (9):

$$\mathbf{o}_{v,m}^{(\ell)} = \underbrace{\sum_{u \in \mathcal{N}_i^{(\ell)}(v)} \alpha_{vu,m}^{(\ell)} \mathbf{v}_{u,m}^{(\ell)}}_{\text{intra-layer message passing}} + \underbrace{\sum_{w \in \mathcal{N}_{inter}^{(\ell)}(v)} \alpha_{vw,m}^{(\ell)} \mathbf{v}_{w,m}^{(\ell)}}_{\text{cross-layer message passing}} \quad (9)$$

Subsequently, the outputs from all attention heads are concatenated and projected through a linear transformation matrix $\mathbf{W}_O^{(\ell)}$. Layer normalization is applied to the projected output, followed by the ELU activation function. For levels where the input and output dimensions match, a residual connection adds the input representation from the previous level to the activated output:

$$\mathbf{h}_v^{(\ell)} = \text{ELU} \left(\text{LayerNorm} \left(\mathbf{W}_O^{(\ell)} \left[\mathbf{o}_{v,1}^{(\ell)} \parallel \dots \parallel \mathbf{o}_{v,H}^{(\ell)} \right] \right) \right) + \mathbb{I}_{d_{in}^{(\ell)}=d_{out}^{(\ell)}} \cdot \mathbf{h}_v^{(\ell-1)} \quad (10)$$

where $[\parallel \dots \parallel]$ denotes the concatenation operation, and $\mathbb{I}_{d_{in}^{(\ell)}=d_{out}^{(\ell)}}$ is an indicator function that equals 1 when input and output dimensions match. The first level transforms the input dimension $d = 70$ to the hidden dimension $d_h \cdot H$ without residual connection due to dimension mismatch.

For the final encoding level $\ell = N_L$, a single-head attention mechanism is employed to reduce the dimensionality from $d_h \cdot H$ to d_h . The activation function and internal residual connection are omitted to preserve the linear separation of the output features:

$$\mathbf{h}_v^{(N_L)} = \text{LayerNorm} \left(\mathbf{W}_O^{(N_L)} \mathbf{o}_{v,1}^{(N_L)} \right) \quad (11)$$

where the single attention head output $\mathbf{o}_{v,1}^{(N_L)} \in \mathbb{R}^{d_h}$ is computed similarly to Eq. (9) but with $H = 1$.

For the initial encoding level ($\ell = 1$), the Graph Transformer directly takes the raw node feature $\mathbf{x}_v$ as input. Simultaneously, a separate linear projection creates a residual pathway to preserve original feature information throughout the depth of the network:

$$\mathbf{x}_v^{res} = \mathbf{W}_{res} \mathbf{x}_v + \mathbf{b}_{res} \quad (12)$$

where $\mathbf{W}_{res} \in \mathbb{R}^{d_h \times d}$ and $\mathbf{b}_{res} \in \mathbb{R}^{d_h}$ project the input features to the final output dimension, and $\mathbf{x}_v^{res}$ is propagated directly to the final aggregation stage via a skip connection.

After N_L encoding levels, the final node embedding is obtained by Eq. (13):

$$\mathbf{h}_v^{final} = \mathbf{h}_v^{(N_L)} + \mathbf{x}_v^{res} \quad (13)$$

Additionally, a graph-level embedding is derived by aggregating all node embeddings through mean pooling to summarize the global network state:

$$\mathbf{g} = \frac{1}{|V|} \sum_{v \in V} \mathbf{h}_v^{final} \quad (14)$$

By projecting these initial structural features into a high-dimensional latent space, the graph transformer encoder effectively resolves potential topological symmetries to extract node-level identities. The multilayer heterogeneous network embedding module outputs both node-level embeddings $\{\mathbf{h}_v^{final}\}_{v \in V}$ and the graph-level embedding $\mathbf{g}$. The procedure is summarized in Algorithm 1.

4.2.2. Dueling QDN module

The Dueling QDN module constitutes the decision-making core of the proposed algorithm. The dueling network separately estimates state value and action advantage functions, then aggregates them to obtain the final Q-values for seed selection.

Traditional QDN estimates the action-value function $Q(s, a)$ directly, which prevents it from distinguishing between the intrinsic value of a state and the relative advantage gained by executing a specific action. The Dueling QDN addresses this limitation by decomposing $Q(s, a)$

Algorithm 1 Multilayer Heterogeneous Network Embedding.

Input: Multilayer heterogeneous network $\mathcal{G}$, current seed sets $\{S_i^{(t)}\}$, covered set $\mathcal{C}$.

Output: Node embeddings $\{\mathbf{h}_v^{final}\}$, graph embedding $\mathbf{g}$.

```

1: for  $i = 1$  to  $L$  do
2:   Compute structural features  $f_v^{(1)}, \dots, f_v^{(S)}$  for all  $v \in V_i$ .
3: end for
4: for each node  $v \in V$  do
5:    $f_v^{cov} \leftarrow \mathbb{I}[v \in \mathcal{C}]$ ,  $f_v^{seed} \leftarrow \mathbb{I}[v \in S_i^{(t)}]$ .
6:    $\mathbf{x}_v \leftarrow [\mathbf{f}_v; f_v^{cov}; f_v^{seed}; \mathbf{1}_{63}] \in \mathbb{R}^{70}$ .
7:    $\mathbf{x}_v^{res} \leftarrow \mathbf{W}_{res} \mathbf{x}_v + \mathbf{b}_{res}$ .
8: end for
9:  $\mathbf{h}_v^{(0)} \leftarrow \mathbf{x}_v$  for all  $v \in V$ .
10: for  $\ell = 1$  to  $N_L$  do
11:   if  $\ell < N_L$  then
12:     Compute  $H$ -head attention outputs  $\{\mathbf{o}_{v,m}^{(\ell)}\}_{m=1}^H$  for all  $v$ .
13:      $\mathbf{z}_v^{(\ell)} \leftarrow \text{LayerNorm} \left( \mathbf{W}_O^{(\ell)} [\mathbf{o}_{v,1}^{(\ell)} \parallel \dots \parallel \mathbf{o}_{v,H}^{(\ell)}] \right)$ .
14:      $\mathbf{a}_v^{(\ell)} \leftarrow \text{ELU}(\mathbf{z}_v^{(\ell)})$ .
15:     if  $d_{in}^{(\ell)} = d_{out}^{(\ell)}$  then
16:        $\mathbf{h}_v^{(\ell)} \leftarrow \mathbf{a}_v^{(\ell)} + \mathbf{h}_v^{(\ell-1)}$ .
17:     else
18:        $\mathbf{h}_v^{(\ell)} \leftarrow \mathbf{a}_v^{(\ell)}$ .
19:     end if
20:   else
21:     Compute single-head attention output  $\mathbf{o}_{v,1}^{(N_L)}$  for all  $v$ .
22:      $\mathbf{h}_v^{(N_L)} \leftarrow \text{LayerNorm} \left( \mathbf{W}_O^{(N_L)} \mathbf{o}_{v,1}^{(N_L)} \right)$ .
23:   end if
24: end for
25:  $\mathbf{h}_v^{final} \leftarrow \mathbf{h}_v^{(N_L)} + \mathbf{x}_v^{res}$  for all  $v \in V$ .
26:  $\mathbf{g} \leftarrow \frac{1}{|V|} \sum_{v \in V} \mathbf{h}_v^{final}$ .
27: return  $\{\mathbf{h}_v^{final}\}_{v \in V}, \mathbf{g}$ .
```

into two separate streams: a state value function $V(s)$ and an action advantage function $A(s, a)$. The state value function $V(s)$ estimates the expected return from being in state s , independent of which action is taken. The component utilizes the graph-level embedding $\mathbf{g}$ to capture the global state information. The value stream is implemented as a two-level multilayer perceptron (MLP) with layer normalization:

$$V(s) = \mathbf{W}_V^{(2)} \cdot \text{ReLU} \left(\text{LayerNorm} \left(\mathbf{W}_V^{(1)} \mathbf{g} + \mathbf{b}_V^{(1)} \right) \right) + \mathbf{b}_V^{(2)} \quad (15)$$

where $\mathbf{W}_V^{(1)} \in \mathbb{R}^{d_{hid} \times d_h}$ and $\mathbf{b}_V^{(1)} \in \mathbb{R}^{d_{hid}}$ are the weight matrix and bias of the first level, $\mathbf{W}_V^{(2)} \in \mathbb{R}^{1 \times d_{hid}}$ and $\mathbf{b}_V^{(2)} \in \mathbb{R}$ are the second level, and d_{hid} denotes the hidden dimension.

The advantage function $A(s, a)$ quantifies the relative benefit of taking action a compared to other available actions in state s . For each candidate node v , the advantage is computed by concatenating the node embedding $\mathbf{h}_v^{final}$ with the graph embedding $\mathbf{g}$. The advantage stream is also implemented as a two-level MLP with layer normalization:

$$A(s, a = v) = \mathbf{W}_A^{(2)} \cdot \text{ReLU} \left(\text{LayerNorm} \left(\mathbf{W}_A^{(1)} [\mathbf{h}_v^{final} \parallel \mathbf{g}] + \mathbf{b}_A^{(1)} \right) \right) + \mathbf{b}_A^{(2)} \quad (16)$$

where $\mathbf{W}_A^{(1)} \in \mathbb{R}^{d_{hid} \times 2d_h}$ and $\mathbf{W}_A^{(2)} \in \mathbb{R}^{1 \times d_{hid}}$ are learnable weight matrices in the first level and the second level, respectively.

The Q-value is computed by aggregating the value and advantage streams. To ensure identifiability and prevent the model from learning arbitrary offsets between $V(s)$ and $A(s, a)$, the mean advantage over all nodes in the network is subtracted:

$$Q(s, a = v; \Theta) = V(s) + \left(A(s, a = v) - \frac{1}{|V|} \sum_{u \in V} A(s, a = u) \right) \quad (17)$$

where V denotes the set of all nodes in the network, and Θ represents all learnable parameters in the model.

4.2.3. Training with the SACRL

The training of SACRL employs Dueling DQN with prioritized experience replay and N-step returns. Firstly, the synthetic three-layer networks are utilized for training and validation, generated by the Erdős-Rényi (ER) network and Barabási-Albert (BA) network. The training dataset comprises a small-scale network with 50 nodes per layer, while the validation dataset is expanded to 500 nodes per layer to evaluate the generalization capabilities. The inter-layer edges between different layers are generated randomly.

Furthermore, training RL agents for influence maximization involves high computational costs for evaluating the influence spread, which requires expensive Monte Carlo simulations. Therefore, a novel proxy reward based on an expected influence estimator is designed. The reward r_t after selecting node v is defined as:

$$r_t = \Phi(S_i^{(t+1)}) - \Phi(S_i^{(t)}) \quad (18)$$

where $S_i^{(t)}$ and $S_i^{(t+1)}$ denote the seed sets of the current target layer i at time steps t and $t+1$, respectively, and $\Phi(\cdot)$ is the expected influence estimator for the target layer. For a given seed set $S_i^{(t)}$ of layer i , the estimator $\Phi(S_i^{(t)})$ combines intra-layer and cross-layer influence estimation:

$$\Phi(S_i^{(t)}) = \text{LIE}_i(S_i^{(t)}) + \sum_{j \neq i} q_{ij} \cdot \Delta_j(A_j) \quad (19)$$

where $\text{LIE}_i(S_i^{(t)})$ estimates the intra-layer influence in layer i , $\Delta_j(A_j)$ estimates the cross-layer contribution to layer j , and q_{ij} is the inter-layer connectivity density. For the intra-layer, the LIE accounts for the two-hop neighbors of the seed set $S_i^{(t)}$, thereby providing a more accurate evaluation of intra-layer influence spread:

$$\text{LIE}_i(S_i^{(t)}) = |S_i^{(t)}| + \left(1 + \frac{\sum_{u \in N_2(S_i^{(t)})} p \cdot d_u^*}{|N_1(S_i^{(t)})|} \right) \cdot \sigma_1^*(S_i^{(t)}) \quad (20)$$

where $N_1(S_i^{(t)})$ denotes the one-hop neighbors of $S_i^{(t)}$ excluding $S_i^{(t)}$ itself, $N_2(S_i^{(t)})$ denotes the two-hop neighbors excluding $S_i^{(t)} \cup N_1(S_i^{(t)})$, d_u^* is the degree of node u within $N_1(S_i^{(t)}) \cup N_2(S_i^{(t)})$, p is the intra-layer propagation probability, and $\sigma_1^*(S_i^{(t)})$ represents the expected number of activated one-hop neighbors:

$$\sigma_1^*(S_i^{(t)}) = \sum_{v \in N_1(S_i^{(t)})} \left(1 - \prod_{u \in S_i^{(t)} \cap \mathcal{N}(v)} (1-p) \right) \quad (21)$$

For the cross-layer, given the current seeds $S_i^{(t)}$ in source layer i , the set of directly reachable nodes in target layer j through inter-layer edges is $A_j = \{v \in V_j : \exists u \in S_i^{(t)}, (u, v) \in E^{inter}\}$. The inter-layer influence contribution to layer j is estimated by Eq. (22):

$$\Delta_j(A_j) = \sum_{v \in N_1(A_j) \setminus A_j} (1 - (1-p)^{r_v}) \quad (22)$$

where $r_v = |\{u \in A_j : (u, v) \in E_j\}|$ is the number of edges connecting v to the activated set A_j . This formulation focuses on the incremental activation of immediate neighbors, which provides a computationally efficient estimate while capturing the cross-layer propagation dynamics. The inter-layer connectivity density q_{ij} is defined as Eq. (2).

Subsequently, during the process of training the model, in order to further enhance the training stability and sample efficiency, the N-step reward mechanism and the prioritized experience replay strategy are incorporated. To ensure stable temporal difference learning, an N-step

return mechanism is employed. The agent accumulates transitions in a buffer and computes the cumulative discounted reward by Eq. (23):

$$R_t^{(n)} = \sum_{l=0}^{n-1} \gamma^l r_{t+l} \quad (23)$$

where n is the multi-step horizon. And the target Q-value for training is computed as Eq. (24):

$$y_t = R_t^{(n)} + \gamma^n \cdot \max_{v \in V_i \setminus S_i^{(t')}} Q(s_{t'}, v; \Theta') \quad (24)$$

where $s_{t'}$ denotes the state observed after n steps, and Θ' denotes the target network parameters. The discount factor γ uses the fixed N-step horizon n for consistency across all transitions. The target network is synchronized with the online network every C updates to stabilize training.

Moreover, prioritized experience replay is employed for sample utilization. Transitions with larger TD-errors are sampled more frequently with probability $P(j) \propto (|\delta_j| + \epsilon)^{\alpha_r}$, where $\delta_j = |y_j - Q(s_j, a_j; \Theta)|$ is the TD-error and α_r controls the prioritization strength. To correct the bias introduced by non-uniform sampling, importance sampling weights are applied:

$$w_j = \left(\frac{1}{|D| \cdot P(j)} \right)^\beta \quad (25)$$

where $|D|$ is the replay buffer size and β is annealed from an initial value β_0 to 1 during training. The weights are further normalized by dividing by the maximum weight for numerical stability. The network parameters Θ are optimized by minimizing the following weighted Huber loss using the Adam optimizer with gradient clipping:

$$\mathcal{L}(\Theta) = \frac{1}{|B|} \sum_{j \in B} w_j \cdot \mathcal{L}_{\text{Huber}}(y_j - Q(s_j, a_j; \Theta)) \quad (26)$$

The complete training procedure is summarized in Algorithm 2. For each episode, a multilayer heterogeneous network is sampled from the training set, and a target layer is randomly selected. This random layer selection strategy enables the model to learn a layer-agnostic policy that generalizes across different network layers. The agent interacts with the environment using the ϵ -greedy policy, and transitions are processed through the N-step buffer before being stored in the prioritized replay memory. The model is periodically evaluated on the validation set. To ensure the assessment of the learned policy, the evaluation is performed using 2000 Monte Carlo simulations. The best-performing model based on influence spread is saved.

4.2.4. Seed selection module

During the training phase, the model updates its parameters to approximate the optimal decision-making policy. In each episode, the target layer is selected via stochastic sampling instead of a deterministic fixed sequence. This stochastic approach provides the necessary state diversity to prevent the agent from overfitting to specific layer orderings, thereby ensuring the learning of a truly layer-agnostic policy that generalizes across heterogeneous network structures. The learned parameters effectively encode the strategy for identifying the most influential seeds. In the inference phase, the trained model is deployed to identify seed nodes in the real-world multilayer heterogeneous network. The selection is performed layer by layer using a greedy policy. Specifically, for each layer i , the agent iteratively selects the node with the highest Q-value until the budget k is exhausted. After each selection, the state is updated, necessitating the recalculation of embeddings and Q-values for the remaining nodes. The complete pseudocode is shown in Algorithm 3.

4.3. Complexity analysis

The computational complexity of SACRL is analyzed according to its modules. In the multilayer heterogeneous network embedding module, computing the five structural centrality features for all nodes costs $O(T \cdot |E_i|)$ per layer, where T denotes the number of iterations until

Algorithm 2 SACRL Training Process.

Input: Training set $\mathcal{T}$, validation set $\mathcal{V}$, budget k , episodes M
Output: Trained parameters Θ^*

- 1: Initialize online network Θ , target network $\Theta' \leftarrow \Theta$.
- 2: Initialize replay memory $\mathcal{D}$, N-step buffer $\mathcal{B}_n$, best score $\sigma^* \leftarrow -\infty$, number of updates $\leftarrow 0$.
- 3: **for** episode $e = 1$ to M **do**
- 4: Sample network $\mathcal{G} \sim \mathcal{T}$, randomly select target layer i .
- 5: Initialize state s_0 with empty seed set $S_i^{(0)} \leftarrow \emptyset$.
- 6: **for** $t = 0$ to $k - 1$ **do**
- 7: Compute Q-values $Q(s_t, v; \Theta)$ for all $v \in V_i \setminus S_i^{(t)}$.
- 8: Select action a_t using ϵ_t -greedy policy based on $Q(s_t, v; \Theta)$.
- 9: Execute action a_t , receive reward r_t , and transition to state s_{t+1} .
- 10: Append (s_t, a_t, r_t) to $\mathcal{B}_n$.
- 11: **if** $|\mathcal{B}_n| \geq n$ or $t = k - 1$ **then**
- 12: Compute $R^{(n)}$ by Eq. (23), push transition to $\mathcal{D}$.
- 13: **end if**
- 14: **if** $|\mathcal{D}| \geq |\mathcal{B}|$ **then**
- 15: Sample batch from $\mathcal{D}$, compute loss by Eq. (26), update Θ .
- 16: Update priorities in $\mathcal{D}$ based on TD-errors.
- 17: number of updates $\leftarrow$ number of updates + 1.
- 18: **end if**
- 19: **if** number of updates $\bmod C = 0$ **then**
- 20: $\Theta' \leftarrow \Theta$.
- 21: **end if**
- 22: Update ϵ_t .
- 23: **end for**
- 24: **if** $e \bmod E_{val} = 0$ **then**
- 25: Evaluate the influence spread on $\mathcal{V}$ via Monte Carlo simulations under the MHIC.
- 26: **if** $\bar{\sigma} > \sigma^*$ **then**
- 27: $\sigma^* \leftarrow \bar{\sigma}$, save $\Theta^* \leftarrow \Theta$.
- 28: **end if**
- 29: **end if**
- 30: $e \leftarrow e + 1$
- 31: **end for**
- 32: **return** Θ^*

Algorithm 3 Seed Selection for MHIM.

Input: Target network $\mathcal{G} = \{G_1, \dots, G_L\}$, budget k , trained parameters Θ^* .
Output: Seed sets $S^* = \{S_1^*, \dots, S_L^*\}$.

- 1: **for** layer $i = 1$ to L **do**
- 2: Initialize $S_i^{(0)} \leftarrow \emptyset$, reset state s_0 .
- 3: **for** $t = 1$ to k **do**
- 4: Compute embeddings by Algorithm 1.
- 5: Compute $Q(s_{t-1}, v; \Theta^*)$ for all $v \in V_i \setminus S_i^{(t-1)}$.
- 6: $v^* \leftarrow \arg \max_{v \in V_i \setminus S_i^{(t-1)}} Q(s_{t-1}, v; \Theta^*)$.
- 7: $S_i^{(t)} \leftarrow S_i^{(t-1)} \cup \{v^*\}$.
- 8: **end for**
- 9: $S_i^* \leftarrow S_i^{(k)}$.
- 10: **end for**
- 11: **return** $S^* = \{S_1^*, \dots, S_L^*\}$

convergence and $|E_i|$ denotes the number of intra-layer edges in layer i . The subsequent graph transformer encoder with cross-layer message passing costs $O(N_L \cdot |E| \cdot d_h \cdot H)$ per forward pass, where N_L is the number of encoding levels, $|E|$ is the total edge count including both intra-layer and inter-layer edges, d_h is the hidden dimension per attention head, and H is the number of attention heads. In the Dueling DQN module, evaluating the state value $V(s)$ and the node-level advantage $A(s, a)$ via two-level MLPs costs $O(|V| \cdot d_{\text{hid}})$ per step, where $|V|$ is the total number of nodes and d_{hid} is the MLP hidden dimension. The proxy reward computation via the LIE estimator and cross-layer contribution Δ_j costs $O(L \cdot (|E_i| + k \cdot |V_i|))$ per step, where L is the number of layers, $|E_i|$ and $|V_i|$ are the edge and node counts of each layer, and k is the seed budget. Aggregating across all modules, the overall time complexity of SACRL is $O(T \cdot L \cdot |E_i| + N_L \cdot |E| \cdot d_h \cdot H + L \cdot (|E_i| + k \cdot |V_i|))$. Regarding space complexity, the node feature matrix, sparse adjacency structures, prioritized replay buffer of capacity C , and model parameters contribute $O(|V| \cdot d)$, $O(|E|)$, $O(C \cdot |V| \cdot d_h)$, and $O(N_L \cdot d_h^2 \cdot H)$ respectively, where d is the input feature dimension, yielding an overall space complexity of $O(C \cdot |V| \cdot d_h + |E| + N_L \cdot d_h^2 \cdot H)$.

5. Experimental results and analysis

5.1. Experimental settings

The hyperparameter configurations for the implementation of the proposed SACRL framework are specified as follows. The Adam optimizer is utilized for SACRL training with a learning rate of 5×10^{-5} and a discount factor $\gamma = 0.99$. The prioritized experience replay buffer is configured with a capacity of 20,000, and the exploration rate ϵ is decayed linearly from 1.0 to 0.05 over the training epochs. Both the value and advantage streams in the Dueling DQN architecture consist of two fully connected layers with 128 hidden units each, incorporating LayerNorm and ELU activation functions.

5.2. Datasets and baselines

Six real-world multilayer heterogeneous networks were included to illustrate the effectiveness of the proposed SACRL. In the six networks, the Arxiv network is a collaboration network where each layer represents a different subject category in the preprint archive. The Aminer network is a multilayer heterogeneous collaboration network consisting of top conferences, whose edges represent author connections. The Paris transportation network is a multilayer network where each layer corresponds to a distinct transportation mode, interconnected into an integrated system. The last three networks are multilayer networks built upon intra-university Facebook social data, where each layer corresponds to a distinct academic cohort and captures intra-cohort social relationships. The inter-cohort connections are modeled as inter-layer edges. Table 1 provides the basic information of these datasets.

To evaluate the effectiveness of the proposed SACRL, the experiments are conducted against the following nine baseline algorithms:

- **CIM** [41]: CIM applies the maximal clique detection strategy to the multilayer network to identify the seed set that maximizes the information spread.
- **CPR** [12]: CPR extends the PageRank algorithm to multilayer networks to characterize the importance of nodes through an information feedback mechanism.
- **CWIIIIF** [30]: CWIIIIF integrates the K-shell and betweenness centrality methods into the multilayer network to identify seed nodes.
- **AGNN** [50]: AGNN employs the autoencoder and graph neural network architectures on complex networks to identify the vital nodes in the network.
- **BiGDN** [27]: BiGDN combines deep reinforcement learning with graph neural networks within an end-to-end model to identify the most influential seed nodes in social networks.

Table 1
The detailed information of multilayer heterogeneous network datasets.

Dataset	Layer	Nodes	Intra-layer Edges	Inter-layer Edges	Type
Aminer	0	23,128	52,067	143,691	Collaboration
	1	28,555	75,464	104,969	
	2	22,102	52,287	116,445	
	3	33,363	84,549	167,437	
	4	24,899	69,997	54,562	
Arxiv	0	1163	2418	2922	Collaboration
	1	4220	12,848	8062	
	2	2230	5019	5152	
	3	201	329	456	
	4	1262	3868	3492	
	5	2682	6272	5947	
	6	1095	2146	3005	
	7	1317	3200	2653	
	8	367	562	1003	
	9	276	588	377	
	10	724	1278	892	
	11	3488	9730	6707	
	12	285	399	808	
BC17	0	287	1389	422	Social
	1	289	1563	463	
	2	284	1310	201	
Cal65	0	775	5699	4240	Social
	1	797	8632	6026	
	2	759	4581	2762	
MU78	0	287	717	1174	Social
	1	282	880	1595	
	2	287	978	1481	
	3	287	870	1031	
	4	278	783	245	
Paris	0	303	356	64	Transportation
	1	241	244	64	

- **CELF** [23]: CELF uses lazy-forward greedy selection with submodular marginal gain to maximize influence spread in multilayer networks.
- **GEHC** [51]: GEHC combines the DeepWalk method with degree and K-shell hybrid centrality, and then calculates node influence by measuring weighted embedding similarities between neighboring nodes.

- **DC**: Degree centrality computes the degree of each node in multilayer networks and selects seed nodes based on the degree scores.
- **K-shell**: K-shell decomposes the network and ranks nodes by coreness to identify influential seeds.

5.3. Ablation experiments

To validate the effectiveness of the structural feature extraction, Dueling DQN architecture, and cross-layer message passing mechanisms in the proposed SACRL algorithm, ablation experiments were conducted on six multilayer heterogeneous networks. Five ablation variants were designed: (1) SACRL-NS, which removes the structural feature extraction, (2) SACRL-DQN, which replaces the proposed Dueling DQN architecture with a standard DQN, (3) SACRL-NC, which eliminates the cross-layer message passing mechanism, (4) SACRL-CD, which augments the node feature vector with a normalized inter-layer degree feature to explicitly encode cross-layer connectivity, and (5) SACRL-CV, which removes the 63-dimensional constant vector from the node feature input. In the ablation studies, the intra-layer spread probability was fixed at $p_{intra} = 0.01$, the seed set sizes were varied between 10 and 50 using a step size of 10, and the Monte Carlo simulations were performed 2000 times. The results of the ablation studies are illustrated in Fig. 4.

As shown in Fig. 4, SACRL consistently outperforms SACRL-NS, SACRL-DQN, and SACRL-NC across all six multilayer heterogeneous networks, validating the contribution of each core component. When $k = 50$, the SACRL demonstrates substantial improvements over SACRL-NS. Specifically, the improvement rates are approximately 26.63% on the Cal65 network and 32.25% on the Aminer network. By integrating both local and global structural features, the SACRL encodes the topological importance of nodes within each layer. The multi-perspective features can be used to generate higher-quality node embeddings. The performance gains are particularly pronounced in Fig. 4(d), where SACRL achieves a 31.46% improvement over the DQN variant at $k = 50$. By decomposing the action-value function into separate state-value and advantage streams, the dueling architecture enables the network to learn the intrinsic value of states and evaluate the relative importance of actions. This leads to the identification of higher-quality seed sets in the multilayer heterogeneous network. Similarly, on the Arxiv network in Fig. 4(a), SACRL outperforms SACRL-NC by 13.03% when $k = 50$. The result indicates that the cross-layer message passing mechanism plays

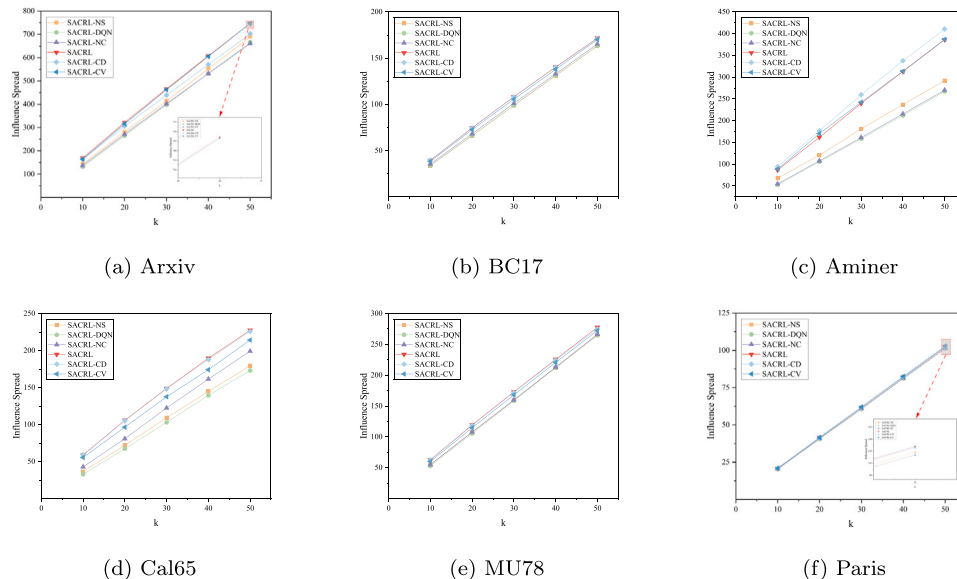


Fig. 4. Performance comparison between different ablation algorithms on the six multilayer heterogeneous networks ($p_{intra} = 0.01$).

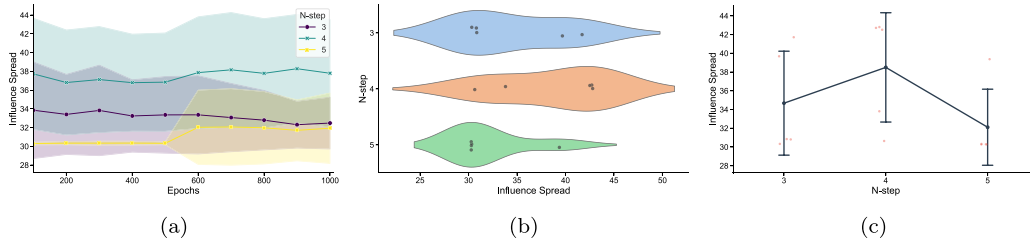


Fig. 5. The selection of the N-step parameter. (a) training process under different N-step settings. (b) Violin plots illustrate distribution of influence spread. (c) statistical comparison of mean performance.

a crucial role in capturing the inter-layer dependencies in multilayer heterogeneous networks with heterogeneous topological structures.

To evaluate the effect of cross-layer degree features, SACRL is compared with SACRL-CD. SACRL outperforms SACRL-CD on the Arxiv network at $k=50$, with the notable improvement of 6.36%. On the four multilayer heterogeneous networks (BC17, MU78, Cal65, and Paris), the performance difference between SACRL and SACRL-CD is marginal, indicating that the graph transformer's cross-layer message passing mechanism already captures the inter-layer structural information through attention-weighted aggregation over inter-layer neighbors, rendering the inter-layer degree feature largely redundant. However, on the large-scale Aminer network, which features denser inter-layer connectivity, SACRL-CD achieves a higher influence spread due to the richer cross-layer structural signals. This suggests that explicitly encoding inter-layer degree is beneficial when inter-layer connections are exceptionally dense. Therefore, for the case where the inter-layer edges are sparse, SACRL's implicit cross-layer message passing provides sufficient and generalizable representations. On the contrary, for cases where there are dense inter-layer connections, incorporating cross-layer degrees as an input feature further enhances the performance of SACRL.

To validate the role of the 63-dimensional constant vector, SACRL is compared with SACRL-CV. SACRL outperforms SACRL-CV on most networks, with the most improvements of 6.19% on the Cal65 network and 1.95% on the MU78 network at $k=50$. The magnitude of the performance gap correlates with the total structural complexity of each network, measured by the average total edge count per node across both intra-layer and inter-layer edges: Cal65 (10.91), BC17 (5.59), MU78 (4.92), Aminer (4.76), Arxiv (3.59), and Paris (1.22). In networks with higher total edge density, node centrality values exhibit a wider distribution, resulting in many structurally peripheral nodes with near-zero feature entries. Without the constant vector, the ELU pre-activations driven by such sparse inputs fall into the low-gradient regime, impeding parameter updates for peripheral nodes. The constant vector provides a learnable bias with 63 independent parameter columns, shifting pre-activations into the effective gradient region of ELU and ensuring non-zero gradient updates for all nodes regardless of their structural centrality. On large-scale networks, the performance difference is negligible, as the constant vector's contribution is diluted during inference when Q-values are ranked across thousands of nodes, and its primary role lies in stabilizing training on small synthetic graphs.

The ablation studies demonstrate that the performance of SACRL is enhanced by integrating the structural feature extraction, Dueling DQN architecture, and cross-layer message passing mechanisms. The comprehensive integration is essential for optimizing influence spread in multilayer heterogeneous networks with diverse topological characteristics.

5.4. Parameter configurations

The sensitivity of SACRL to the N-step return horizon N was investigated through parameter experiments. Following the DRL-based work for influence maximization [27], k is fixed at 5, as larger values have been shown to cause significant computational training overhead with

negligible performance gain. The model is trained for 1000 epochs on ER multilayer heterogeneous networks consisting of 50 nodes per layer, with N set to 3, 4, and 5, respectively. Each configuration is executed five times independently, and the influence spread is evaluated every 100 epochs on the BA network comprising 500 nodes per layer. The results are presented in Fig. 5.

As shown in Fig. 5(a), the curve with $N=4$ consistently achieves the highest mean influence spread, while the curve with $N=3$ starts at a comparable level and exhibits a slight downward trend in the later training stages. Fig. 5(b) further illustrates the distribution of influence spread across all evaluation checkpoints, where $N=4$ shows a greater density toward higher values compared to the other two configurations. The mean influence spread reaches its maximum at $N=4$, followed by $N=3$ and $N=5$, as illustrated in Fig. 5(c). Moreover, all three configurations exhibit the variability across runs, which is expected given the combinatorial nature of the MHIM problem. The complex inter-layer link patterns in multilayer heterogeneous networks create multiple local optima in the solution space. Furthermore, the cascading effect across layers amplifies the uncertainty in influence estimation, as the activation of a seed node propagates through both intra-layer and inter-layer edges, leading to diverse diffusion outcomes. Additionally, as the seed set expands, the coverage regions of high-value nodes gradually overlap, resulting in diminishing marginal gains that reduce the discriminability of subsequent decisions and increase the difficulty of policy learning.

Despite the variability observed during the 1000 training epochs, $N=4$ achieves a distinctly higher mean influence spread, marking the configuration as a highly effective trade-off for the MHIM problem. The setting $N=4$ is therefore adopted as the default in all the experiments.

5.5. Performance comparisons with baselines

5.5.1. Comparison on influence spread

To demonstrate the effectiveness of SACRL, nine baseline algorithms were compared: CIM, CPR, CWIIF, AGNN, BiGDN, CELF, GEHC, DC, and K-shell, in terms of influence spread on six multilayer heterogeneous networks under the proposed propagation model. The intra-layer spread probability was fixed at $p_{intra} = 0.01$, the seed set sizes were varied between 10 and 50, and the Monte Carlo simulations were performed for 2000 times. Fig. 6 illustrates the influence spread when $p_{intra} = 0.01$.

By analyzing the results in Fig. 6, it is evident that SACRL outperforms all algorithms except the CELF in most cases. As shown in Fig. 6(a)–(f), the proposed SACRL achieves superior performance in influence propagation across six multi-layer heterogeneous networks with diverse scales and topological structures. In Fig. 6(a), when $k=50$, SACRL achieves an influence spread closely approaching that of CELF while outperforming all other baselines. On the contrary, CWIIF performs poorly on all multilayer heterogeneous networks. The reason why CWIIF performs the worst on all networks is that CWIIF selects a unified seed set throughout the entire network, without being able to distinguish the importance of nodes at different layers in multilayer heterogeneous networks.

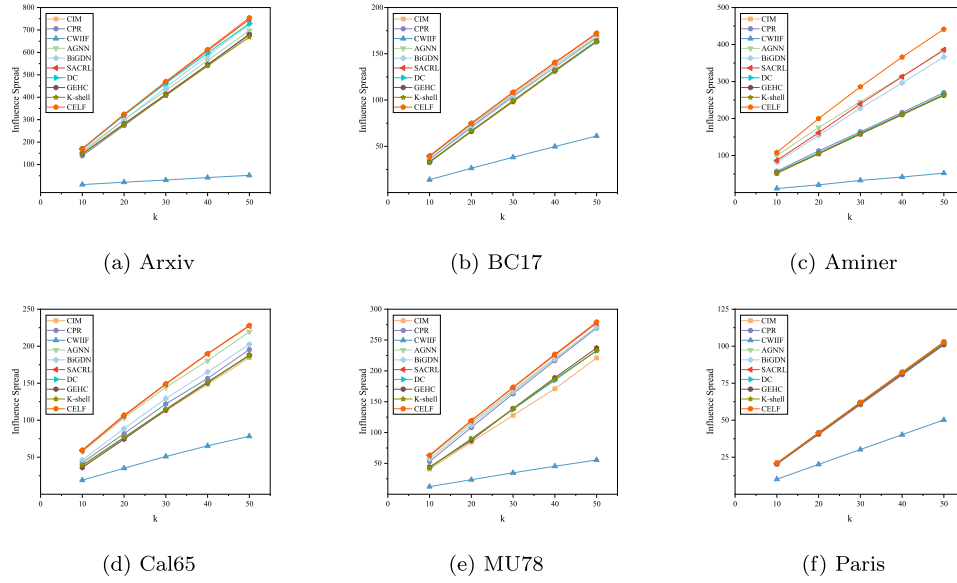


Fig. 6. Comparison on influence spread between ten algorithms on the six multilayer heterogeneous networks ($p_{intra} = 0.01$).

As shown in Fig. 6, CELF consistently achieves the highest influence spread across all six multilayer heterogeneous networks, serving as an approximate upper bound owing to its exhaustive greedy evaluation via Monte Carlo simulations. On the five networks (Arxiv, BC17, MU78, Cal65, and Paris), the influence spread of SACRL is close to that of CELF, demonstrating near-optimal performance. On the large-scale Aminer network, CELF outperforms SACRL by 14.34% at $k = 50$. This larger gap can be attributed to the scale and dense inter-layer connectivity of Aminer, where the MHIC model produces complex cascading dynamics that are difficult to approximate with the proxy reward. In contrast, CELF's direct Monte Carlo evaluation captures global influence spread but at a higher computational cost, which becomes prohibitive at scale. SACRL bridges this trade-off by learning a generalizable policy that approximates near-optimal selection with significantly lower inference overhead.

On the Cal65 network in Fig. 6(d), compared with the second-best algorithm AGNN, SACRL still maintains a 3.58% improvement. In Fig. 6(e), SACRL achieves approximately 277 at $k = 50$, outperforming AGNN by 2.5% and BiGDN by 2.2%. Although AGNN and BiGDN capture intra-layer structural importance in multilayer heterogeneous networks, they face limitations in addressing the inter-layer interactions inherent to real-world multilayer systems. Specifically, since AGNN and BiGDN focus on single-layer or aggregated topological feature extraction, they fail to fully consider the impact of many-to-many cross-layer connections on node importance. In such multilayer heterogeneous networks, the inter-layer interaction is fundamental to identifying key spreaders. The performance gain is more pronounced on networks with richer inter-layer connections, such as the Arxiv and Cal65 networks, where modeling cross-layer interactions is critical and SACRL captures the intricate cross-layer topological structures more effectively than AGNN or BiGDN.

The magnitude of SACRL's improvement over AGNN and BiGDN varies across datasets and is related to inter-layer structural complexity. On the Aminer network, which features a large scale and dense inter-layer connections, AGNN's autoencoder-based architecture can effectively extract single-layer features and identify high-quality seed nodes, which maximizes intra-layer propagation to create more opportunities for cross-layer spread. Similarly, on the Paris network, the minimal inter-layer coupling suppresses the effect of cross-layer-aware methods, causing all algorithms to converge toward comparable performance.

In contrast to machine learning-based methods, the traditional centrality-based algorithms DC and K-shell exhibit weaker performance on most networks. At $k = 50$, DC achieves only 82.36% of SACRL's influence spread on the Cal65 network, while K-shell achieves 82.51% on the same network. GEHC, which augments degree and K-shell centrality with DeepWalk-based graph embeddings, achieves slightly higher scores than DC and K-shell on several networks but still falls short of SACRL by 17.28% on the Cal65 network and 14.59% on the MU78 network. The performance gap reflects the structural characteristics of these networks: the Cal65 network features high intra-layer density coupled with inter-layer connectivity, creating complex cross-layer propagation pathways that simple degree or coreness rankings fail to capture. The three methods share a common limitation in that they do not explicitly model many-to-many inter-layer connections, leading to seed sets that are locally central but globally suboptimal in multilayer heterogeneous networks. The exception is the Paris network, where the sparse inter-layer connectivity renders the cross-layer topology negligible, allowing local centrality to approximate node importance with reasonable accuracy.

On the Paris network, the performance differences among algorithms are minimal. This convergence is primarily due to the network's extremely sparse inter-layer connectivity, comprising only 64 inter-layer edges, which renders cross-layer propagation negligible and can reduce the problem to near-independent intra-layer spreading. Under such conditions, the topological structure is sufficiently simple that the most critical nodes can be readily identified without sophisticated cross-layer modeling, thereby limiting the advantage of the proposed SACRL. To further compare the effectiveness, Table 2 presents the comparative performance and rankings of all algorithms at $k = 50$. The influence spread rankings of the proposed SACRL and baseline algorithms across different multilayer heterogeneous networks demonstrate that SACRL consistently achieves top-tier performance.

5.5.2. Comparison on the seed set

To provide more insight into the performance of the proposed SACRL, the distribution of the seed nodes in the multilayer heterogeneous network is evaluated by the average distance between seeds. A larger value of the metric indicates that the selected seed nodes are more widely distributed across the network layers, which can reduce

Table 2

Comparative algorithm performance evaluated by relative influence spread at a seed set size of $k = 50$. The performance of the second-ranked algorithm on each dataset serves as the baseline (100%).

Algorithm	Arxiv	BC17	MU78	Cal65	Paris	Aminer	Rank
CELF	100.96	100.39	100.58	100.28	100.20	114.34	1
SACRL	100.00	100.00	100.00	100.00	100.00	100.00	2
AGNN	93.40	98.28	97.57	96.54	98.90	99.23	3
BiGDN	96.97	97.21	97.85	89.12	99.52	94.91	4
CPR	91.33	97.67	97.05	85.98	98.19	69.88	5
DC	97.55	94.85	83.82	82.36	99.86	68.65	6
GEHC	90.89	95.46	85.41	82.72	98.72	68.16	7
K-shell	89.42	95.43	83.80	82.51	99.42	68.18	8
CIM	90.80	94.93	79.59	81.17	98.50	68.04	9
CWIIIF	6.92	35.60	20.01	34.45	48.92	13.57	10

the overlap of influence. The average distance between seeds can be calculated using Eq. (27):

$$L(S) = \frac{2}{k(k-1)} \sum_{u,v \in \bigcup_{i=1}^L S_i, u < v} d_G(u, v) \quad (27)$$

where $d_G(u, v)$ denotes the shortest path between nodes u and v in the multilayer heterogeneous network. The seed set is denoted by $\bigcup_{i=1}^L S_i$ and the size of the seed set is denoted by k .

Fig. 7 shows the average distance between the seeds selected by different algorithms across six multilayer heterogeneous networks. The experimental results demonstrate that the seeds selected by BiGDN and CIM, as well as traditional centrality methods such as DC and K-shell, exhibit higher average distances in most networks. In comparison, the average distance among seeds selected by SACRL remains at a relatively low level across all six multilayer heterogeneous networks. Nevertheless, SACRL achieves superior performance in influence spread. This non-linear relationship between seed distance and performance suggests that a large spatial dispersion is not a prerequisite for effective influence maximization on multilayer heterogeneous networks. While a relatively concentrated distribution might imply potential influence overlap, SACRL avoids the excessive dispersion. For DC and K-shell, the large average distance reflects their tendency to identify hubs within

individual layers that lack inter-layer connectivity, failing to propagate across the multilayer networks.

To further evaluate the spreading capability of seed nodes in multilayer heterogeneous networks, the external cross-layer edges, which represent the inter-layer connections where one endpoint is a seed node and the other is a non-seed node, were analyzed in Fig. 8. The metric measures the potential of seed nodes to spread influence across different layers to unactivated regions. Fig. 8(a) and (b) present the external cross-layer edges of seed nodes selected by ten algorithms across most datasets. This indicates that the seeds selected by SACRL possess superior one-hop cross-layer spreading capability. While BiGDN, DC, and K-shell show larger average distances, their cross-layer edges rank lower, confirming that wide distribution does not necessarily correlate with strong inter-layer connectivity. SACRL's advantage lies in identifying seeds that, despite being more clustered, act as vital cross-layer bridges. Notably, CELF exhibits fewer external cross-layer edges than SACRL on several networks yet achieves the highest influence spread. This shows that external cross-layer edges only measure one-hop direct connections, while CELF uses multi-step simulations to find seeds that can activate more nodes further downstream.

Based on the results shown in Figs. 7 and 8, SACRL demonstrates an effective seed selection strategy by achieving the highest external cross-layer connectivity while maintaining a moderate average distance, which enables efficient influence propagation across both intra-layer and inter-layer pathways. Finally, the findings suggest that effective influence maximization in multilayer heterogeneous networks requires careful consideration of both intra-layer distribution and inter-layer connectivity of seed nodes.

5.6. Graph transformer analysis

To evaluate the necessity of the GT architecture and the rationality of mean pooling, four additional ablation variants are designed: SACRL-GCN replaces the GT encoder with a Graph Convolutional Network. SACRL-MLP eliminates graph-based message passing entirely and applies an MLP to raw node features. SACRL-MP and SACRL-SP substitute max pooling and sum pooling for mean pooling, respectively. The results are shown in Fig. 9.

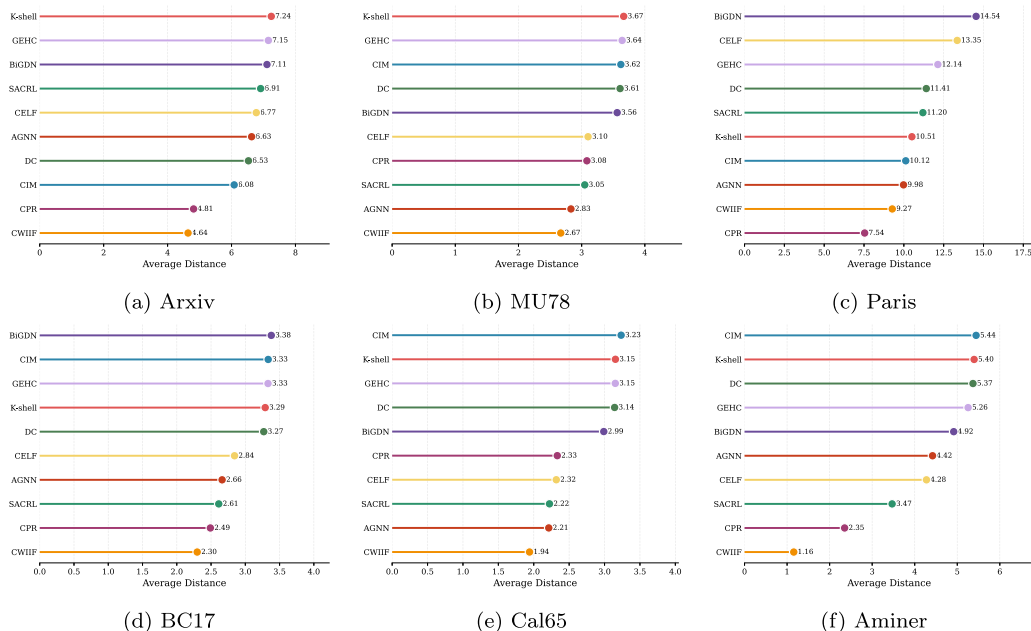


Fig. 7. Average distance between seeds selected by ten algorithms across six multilayer heterogeneous networks.

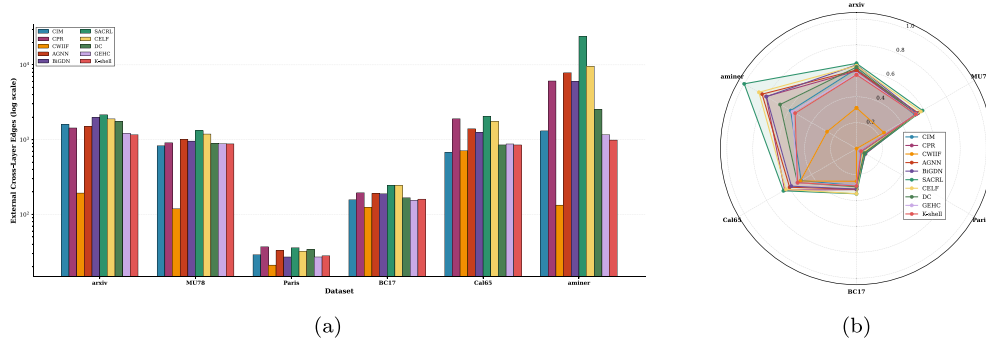


Fig. 8. (a) External cross-layer edges of seed nodes across six multilayer heterogeneous networks. (b) Radar chart comparison of external cross-layer edges across ten algorithms.

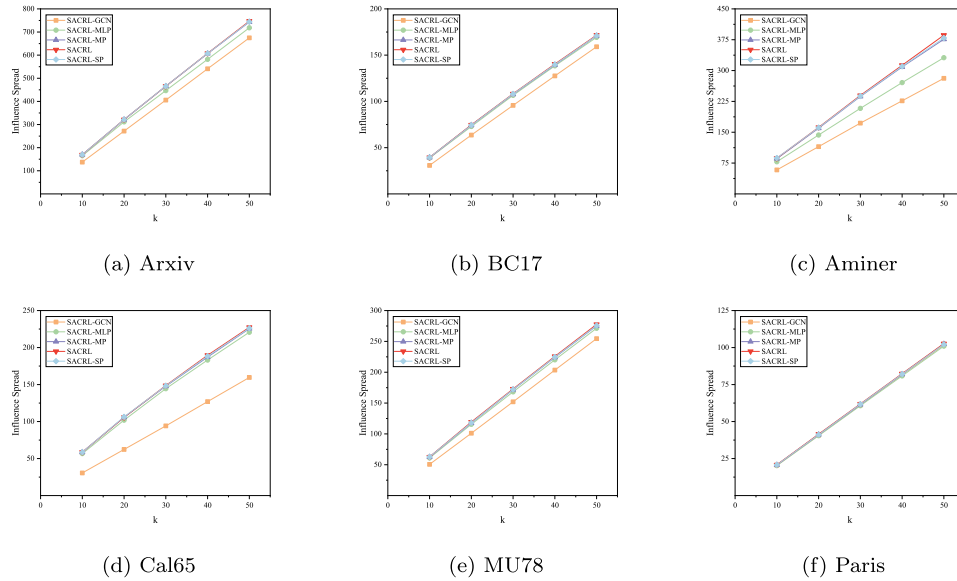


Fig. 9. Performance comparison between different GNN architecture and pooling strategy variants on the six multilayer heterogeneous networks ($p_{intra} = 0.01$).

SACRL consistently outperforms SACRL-GCN and SACRL-MLP across all six networks, confirming that performance gains cannot be attributed solely to handcrafted structural features. When $k = 50$, SACRL achieves improvement rates over SACRL-GCN of 42.44% on the Cal65 network and 37.30% on the Aminer network, and outperforms SACRL-MLP by 16.52% on Aminer network. Since SACRL and SACRL-GCN share identical input features, the gap isolates the contribution of the GT's attention-weighted aggregation over GCN's uniform normalization. In the Cal65 and Aminer networks, GCN's uniform symmetric normalization aggregates neighbor information without discrimination, which may cause over-smoothing of node representations and reduce the discriminability of structurally distinct nodes. The GT's attention mechanism mitigates this issue by assigning differentiated weights to neighbors, thereby preserving structural heterogeneity in the learned embeddings. SACRL-MLP exhibits the overall performance degradation, confirming that topology-aware message passing is indispensable and that structural features alone are insufficient for effective node representation.

Regarding pooling strategy, SACRL with mean pooling achieves the highest influence spread across all six networks, with the performance gap most evident in Fig. 9(c), where SACRL outperforms SACRL-MP by 2.63% and SACRL-SP by 1.83% at $k = 50$. Max pooling retains only the maximum activation per embedding dimension, which is dominated by structurally extreme nodes in dense networks and produces a biased global state representation. Sum pooling accumulates embeddings

across all nodes, causing the graph embedding magnitude to scale with network size. This inconsistency between the small training networks and large inference networks undermines cross-network generalization. Mean pooling normalizes by node count, yielding a scale-invariant global state representation that remains stable across networks of vastly different sizes. On small networks such as the Paris and BC17 networks, all pooling strategies yield comparable results, as the reduced node count limits variance between pooling operations.

The results confirm that the GT architecture contributes beyond handcrafted feature engineering, and that mean pooling provides the most robust and generalizable global state representation for the Dueling DQN across multilayer heterogeneous networks of diverse topological characteristics.

5.7. Comparison on time efficiency

To assess the computational efficiency and scalability of SACRL, the inference time was evaluated across six multilayer heterogeneous networks against CELF as a baseline and against two machine learning-based methods, AGNN and BiGDN. The inference time is defined as the total time required to select seed nodes across all L layers for a given network at seed budget k . Furthermore, the resource consumption of SACRL was measured as the peak memory usage during the inference process. Table 3 shows the training time on synthetic networks.

Table 3

Training time comparison between SACRL and SACRL-MC (8000 times) on synthetic networks.

Algorithm	Proxy Reward	Training Time (s)
SACRL	Yes	582.85
SACRL-MC	No (Full MC)	1216.74

Fig. 10 presents the inference time at $k = 30$. CELF incurs the highest runtime across all datasets, as its greedy strategy evaluates the marginal gain of every candidate node via thousands of Monte Carlo simulations at each selection step, resulting in a time cost that grows rapidly with both network size and seed budget. AGNN exhibits higher inference time than the two DRL-based methods. This is because AGNN adopts an autoencoder architecture: when applied to a new target network, the autoencoder must be re-executed on the new graph to generate node embeddings, precluding the amortization of representation learning across different networks. In contrast, BiGDN is a DRL-based method whose trained parameters generalize directly to unseen networks, and inference reduces to a single forward pass. SACRL requires recomputing the node embeddings after seed selection. While the overhead is negligible on small networks, it incurs additional time on large-scale networks, resulting in lower time efficiency compared to BiGDN. However, SACRL significantly outperforms BiGDN on influence spread.

Table 4 further reports SACRL's inference time, Monte Carlo evaluation time, peak GPU memory, and peak CPU memory for $k \in \{10, 20, 30, 40, 50\}$. The inference time increases approximately linearly with k , as each additional seed selection requires a recomputation of embeddings for the remaining candidate nodes. Peak memory remains constant across all values of k for a given dataset since the graph structure is fixed during inference, and scales with network size.

5.8. Generalization and proxy reward analysis

5.8.1. Distribution shift and generalization analysis

SACRL was trained on synthetic ER networks (50 nodes per layer) but tested on real-world networks with different topological characteristics. To quantify this generalization gap, the topological statistics and distributional distances between training and test networks were computed.

Table 5 reports the key topological statistics. The training networks have low clustering coefficients, while real networks exhibit higher clustering coefficients. The average degree difference is marginal, but the skewness gap and clustering gap are significant, confirming that the distributional shift lies primarily in degree heterogeneity and community structure.

Table 4

Resource consumption of SACRL across six multilayer heterogeneous networks for varying seed budget k . Infer. Time: seed selection time (s). MC Time: Monte Carlo evaluation time (s). Peak GPU and Peak CPU: maximum memory usage (MB).

Dataset	k	Infer. Time (s)	MC Time (s)	Peak GPU (MB)	Peak CPU (MB)
BC17	10	0.5440	0.0995	89.38	0.24
	20	0.0523	0.1528	89.38	0.20
	30	0.0780	0.2033	89.38	0.20
	40	0.0971	0.2477	89.38	0.20
	50	0.1261	0.2732	89.38	0.20
Cal65	10	0.4051	0.3407	421.89	0.92
	20	0.2291	0.5466	421.89	0.97
	30	0.3344	0.7342	421.89	0.97
	40	0.4423	0.8956	421.89	0.97
	50	0.5376	1.0097	421.89	0.97
MU78	10	0.0780	0.1776	125.74	0.40
	20	0.1550	0.2787	125.74	0.39
	30	0.2795	0.3562	125.74	0.38
	40	0.3420	0.4112	125.74	0.38
	50	0.4221	0.4596	125.74	0.38
Aminer	10	34.9576	4.5044	10,116.68	41.52
	20	68.6549	6.0012	10,116.68	42.08
	30	101.8899	6.8074	10,116.68	42.23
	40	134.5600	9.3769	10,116.68	42.32
	50	175.4090	12.6278	10,116.68	42.46
Arxiv	10	1.7563	0.6022	1170.68	3.16
	20	3.0015	1.1187	1170.07	3.26
	30	4.5979	1.2524	1170.07	3.74
	40	6.0171	1.6080	1170.07	3.80
	50	7.5224	1.6650	1170.07	3.85
Paris	10	0.0120	0.0150	23.15	0.17
	20	0.0235	0.0220	23.08	0.17
	30	0.0320	0.0310	23.08	0.17
	40	0.0440	0.0387	23.08	0.17
	50	0.0832	0.0445	23.08	0.17

Table 5

Topological statistics of training and test networks. $\bar{d}$: average degree; γ_d : degree skewness; CC: average clustering coefficient.

Network	Type	Nodes	$\bar{d}$	γ_d	CC
Synthetic ER (avg.)	Train	150	7.19	0.17	0.1478
Arxiv	Test	19,310	5.04	4.99	0.7375
BC17	Test	860	10.00	1.37	0.2036
Cal65	Test	2331	16.48	1.49	0.2388
MU78	Test	1421	6.29	1.68	0.1992
Paris	Test	544	2.21	2.68	0.0049
Aminer	Test	132,047	4.84	6.73	0.7165

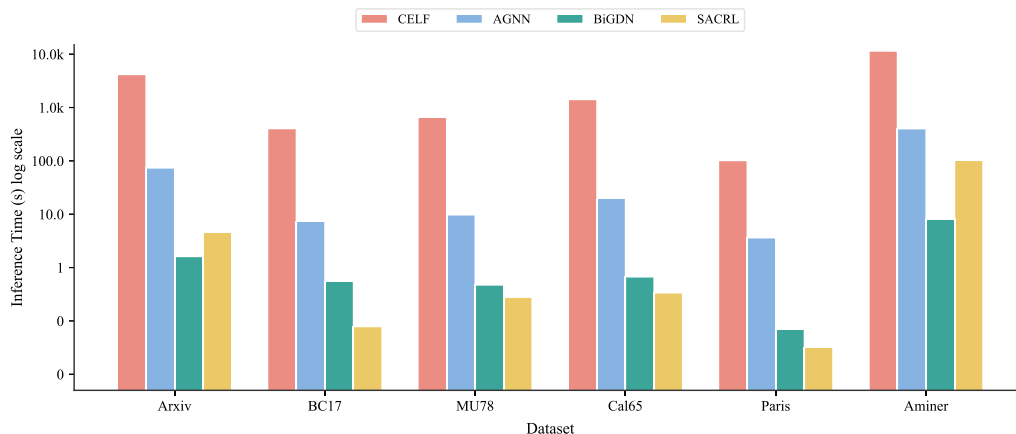


Fig. 10. Inference time comparison at $k = 30$ of SACRL against CELF, AGNN, and BiGDN across six multilayer heterogeneous networks.

Table 6
KS test statistics between training and test distributions. All $p < 0.001$.

Network	D_{deg}	D_{CC}
Arxiv	0.4957	0.7670
BC17	0.2925	0.1753
Cal65	0.4941	0.3083
MU78	0.2879	0.1895
Paris	0.8481	0.8579
Aminer	0.5301	0.7367

Table 6 presents the two-sample Kolmogorov–Smirnov (KS) test results. All p -values are below 0.001, confirming statistically significant distributional shifts across all training–test pairs. The Paris network exhibits the largest shift ($D_{\text{deg}} = 0.8481$, $D_{\text{CC}} = 0.8579$) due to its sparse topology, while BC17 and MU78 show relatively moderate shifts ($D_{\text{deg}} < 0.30$, $D_{\text{CC}} < 0.20$).

Despite the distributional gaps, SACRL ranks second only to CELF on all six networks. The generalization stems from three factors: (1) the layer-wise feature normalization provides scale-invariant representations, (2) the attention-weighted cross-layer message passing adapts to local topology, and (3) the random layer selection during training promotes a layer-agnostic policy.

To examine whether a more realistic graph generator improves generalization, SACRL-LFR is trained using LFR benchmark networks with planted communities. Table 7 presents the results. SACRL outperforms SACRL-LFR on five of six datasets across all seed budgets. SACRL-LFR is marginally better only on Aminer at $k \leq 20$, where the LFR community structure provides an initial advantage for identifying locally influential seeds. However, as k increases, ER-trained SACRL surpasses SACRL-LFR from $k = 30$ onward. This result suggests that the structural simplicity of ER networks forces the agent to rely on learned features rather

than memorizing specific community patterns, thereby yielding more generalizable representations.

5.8.2. Proxy reward validation

To validate the accuracy of the proposed proxy reward function, correlation analysis is conducted between the proxy estimates and the true influence spread measured by Monte Carlo simulation. For each test network and layer, 50 random seed sets of size $k = 5$ are generated, and both the proxy estimate $\Phi(S)$ and the true MC influence $\sigma(S)$ with 1000 simulation runs are computed. The Spearman rank correlation (ρ) and Pearson linear correlation (r) are reported. Fig. 11(a) presents the correlation coefficients across synthetic networks of varying sizes and the BC17 network, and Fig. 11(b) visualizes the linear relationship between proxy and MC estimates on the BC17 network.

All correlations are statistically significant ($p < 0.01$), confirming that the proxy reward consistently preserves the relative ranking of seed set quality. A notable finding is that the correlation coefficients are substantially higher on the real-world BC17 network compared to the small synthetic networks. The proxy reward formula is designed based on structure: the LIE estimator captures intra-layer two-hop activation potential, while the cross-layer component estimates inter-layer influence contribution weighted by connectivity density. These structural approximations align closely with the actual diffusion dynamics, yielding ranking agreement.

The marginal gain ranking accuracy is evaluated by computing the proxy and MC marginal gains for individual nodes selected as the sole seed. On a synthetic network with 40 nodes per layer, the Spearman correlation between proxy and MC marginal gains is 0.502, and the Top-5 overlap rate (i.e., the fraction of the proxy’s top-5 ranked nodes that also appear in the MC’s top-5) is 40%. For reinforcement learning, the critical requirement is that the reward signal preserves the ordinal ranking of actions, not that it provides exact influence estimates. The consistently significant correlations confirm that the proxy reward satisfies this requirement.

Table 7
Influence spread of SACRL (ER training) vs. SACRL-LFR (LFR training). Bold indicates the better result ($p_{\text{intra}} = 0.01$).

Dataset	$k = 10$		$k = 20$		$k = 30$		$k = 40$		$k = 50$	
	SACRL	SACRL-LFR	SACRL	SACRL-LFR	SACRL	SACRL-LFR	SACRL	SACRL-LFR	SACRL	SACRL-LFR
Arxiv	169.43	165.01	321.41	314.15	465.72	457.40	607.89	597.13	747.11	734.04
BC17	39.47	38.89	74.65	73.42	108.25	106.27	140.39	138.24	171.65	169.33
Cal65	58.73	58.20	105.69	102.89	148.67	143.93	189.58	183.28	227.38	221.61
MU78	62.70	62.05	119.36	116.77	172.93	169.76	225.71	221.21	277.59	272.23
Paris	20.82	20.68	41.48	41.24	61.93	61.67	82.38	81.89	102.69	102.18
Aminer	86.76	90.10	161.85	163.12	239.53	233.55	312.91	299.72	386.11	364.22

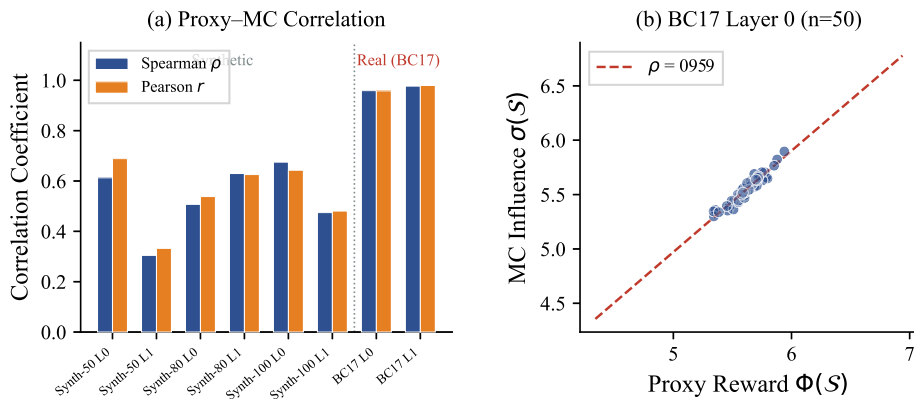


Fig. 11. Proxy reward validation. (a) Spearman and Pearson correlation coefficients between proxy reward estimates and MC simulations across synthetic and real networks (50 seed sets of size $k = 5$ per configuration, all $p < 0.01$). (b) Scatter plot of proxy reward vs. MC influence on BC17 layer 0, demonstrating near-perfect ranking agreement ($\rho = 0.963$).

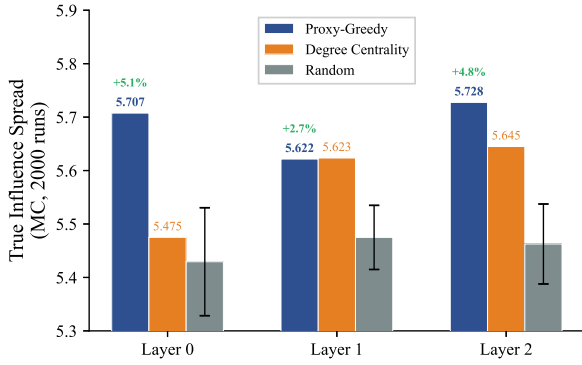


Fig. 12. True influence spread (MC simulation, 2000 runs) of seed sets selected by proxy-trained greedy, degree centrality, and random strategies on a synthetic 3-layer network (60 nodes per layer, $k = 5$). The green annotations indicate the improvement of proxy-greedy over random selection.

To validate that the proxy reward function itself provides an effective optimization signal, we compare seed sets selected by maximizing the proxy marginal gain against degree centrality and random selection. All influence values are measured by true MC simulation. Fig. 12 presents the results on a synthetic 3-layer network with 60 nodes per layer and seed budget $k = 5$. The proxy-greedy strategy consistently outperforms the baselines, confirming that the proxy reward correctly identifies high-quality seeds. Building upon this validated reward function, the full SACRL framework employs Dueling DQN to learn a policy that generalizes across networks.

The greedy strategy guided by the proxy marginal gain consistently achieves the highest true MC influence across all three layers, outperforming random selection by 2.68%–5.12% and degree centrality by up to 3.55%. The improvement over degree centrality, which only considers intra-layer connectivity, highlights that the proxy reward’s cross-layer component successfully identifies seeds with superior cross-layer propagation capability. Although these improvements are observed on a small synthetic network (60 nodes per layer), the performance advantages are more pronounced on real-world networks. The consistent superiority of the SACRL over all baselines under true MC evaluation across all six real-world networks provides the strongest evidence that optimizing the validated proxy reward leads to policies that maximize the correct objective.

5.9. Statistical results

To ensure the statistical validity of the experimental results, the Wilcoxon signed-rank test was performed. The Wilcoxon signed-rank test is a non-parametric statistical method suitable for assessing whether significant differences exist between two comparative algorithms. The significance level was set at $\alpha = 0.05$. The results across the six multilayer heterogeneous networks are summarized in Table 8, where the Z -value measures the magnitude of the performance difference, and the p -value indicates the probability of observing these differences under the null hypothesis. A p -value below 0.05 confirms a statistically significant difference. Additionally, $N+$ represents the number of networks where the proposed SACRL algorithm outperforms the baseline algorithm, while $N-$ indicates the instances where SACRL underperforms.

According to the results in Table 8, SACRL demonstrates statistically significant superiority over CIM, CPR, CWIIF, BiGDN, and GEHC across all evaluated seed budgets. When compared with traditional centrality methods such as DC and K-shell, SACRL maintains a consistent statistical advantage for all k values. In the comparison with AGNN, no significant differences are observed at budgets ranging from $k = 10$ to $k = 40$. However, SACRL exhibits statistically significant superiority at $k = 50$. Conversely, CELF achieves higher influence spread than SACRL, showing

Table 8

The statistical results of the Wilcoxon test on SACRL and the other nine algorithms at $\alpha = 0.05$ on the six multilayer heterogeneous networks.

SACRL vs.	k	$N-$	$N+$	Z	p -value
CIM	10	0	6	-2.201	0.028
	20	0	6	-2.201	0.028
	30	0	6	-2.201	0.028
	40	0	6	-2.201	0.028
	50	0	6	-2.201	0.028
CPR	10	0	6	-2.201	0.028
	20	0	6	-2.201	0.028
	30	0	6	-2.201	0.028
	40	0	6	-2.201	0.028
	50	0	6	-2.201	0.028
CWIIF	10	0	6	-2.201	0.028
	20	0	6	-2.201	0.028
	30	0	6	-2.201	0.028
	40	0	6	-2.201	0.028
	50	0	6	-2.201	0.028
AGNN	10	1	5	-0.943	0.345
	20	1	5	-1.153	0.249
	30	1	5	-1.153	0.249
	40	1	5	-1.728	0.075
	50	0	6	-2.201	0.028
BiGDN	10	0	6	-2.201	0.028
	20	0	6	-2.201	0.028
	30	0	6	-2.201	0.028
	40	0	6	-2.201	0.028
	50	0	6	-2.201	0.028
CELF	10	6	0	-2.201	0.028
	20	5	1	-1.728	0.075
	30	6	0	-2.201	0.028
	40	5	1	-1.992	0.046
	50	6	0	-2.201	0.028
DC	10	1	5	-1.992	0.046
	20	0	6	-2.201	0.028
	30	0	6	-2.201	0.028
	40	0	6	-2.201	0.028
	50	0	6	-2.201	0.028
GEHC	10	0	6	-2.201	0.028
	20	0	6	-2.201	0.028
	30	0	6	-2.201	0.028
	40	0	6	-2.201	0.028
	50	0	6	-2.201	0.028
K-shell	10	1	5	-1.992	0.046
	20	0	6	-2.201	0.028
	30	0	6	-2.201	0.028
	40	0	6	-2.201	0.028
	50	0	6	-2.201	0.028

statistically significant differences at most k values. Overall, the statistical analysis validates the efficacy of the SACRL compared to the majority of the baseline methods.

6. Discussion

The effectiveness of the SACRL algorithm relies on its ability to model the complex topology in multilayer heterogeneous networks. The ablation studies demonstrate that the performance of SACRL is enhanced by integrating the structural feature extraction, Dueling DQN architecture, and cross-layer message passing mechanisms. This integration enables SACRL to capture complex intra-layer and inter-layer topological information in an effective way. Furthermore, the influence spread experimental results further validate the capability of the proposed SACRL algorithm to identify seed sets across a variety of real-world multilayer heterogeneous networks.

However, to balance computational efficiency with training stability, the optimization strategy adopts a proxy reward function to circumvent

the high cost of Monte Carlo simulations. While the efficient estimator is anchored in second-order activation potential and effectively reduces training time, it inherently focuses the agent on local topological saturation. Consequently, a marginal approximation gap may exist between the neighborhood-based approximation and the true global influence spread. To bridge the gap, a promising direction for future research is to explore hybrid reward mechanisms that incorporate periodic global feedback, thereby aligning the learned policy more closely with global optimality while preserving training efficiency.

Furthermore, the current propagation model relies on inter-layer connectivity density to estimate diffusion probabilities and, within a given layer pair (i, j) , assigns a uniform activation probability q_{ij} to all inter-layer edges, thereby abstracting away node-pair-specific connection strengths. While this static formulation is effective for structural analysis, it does not fully capture the dynamic nature of real-world propagation, which is often modulated by temporal shifts and multi-dimensional user attributes. Future research should transcend these static constraints by incorporating fine-grained transmission probabilities informed by interaction frequency or edge weights (when available), as well as dynamic node features, and by extending the approach to multilayer higher-order networks. In addition, extending the current per-layer budget to a global budget with adaptive layer-wise allocation offers another promising direction, as it enables optimal resource distribution across layers under a fixed total budget while respecting the varying influence potential of each layer.

The feasibility of identified seeds under structural damages is also a vital consideration for practical deployment. The SACRL does not account for robustness issues in multilayer heterogeneous networks. Given the correlation between network robustness and influence maximization [52], the integration of structural resilience into the SACRL framework represents a promising future research direction. By incorporating robustness-aware metrics into the reward function, agents could be trained to prioritize seeds that maintain propagation efficacy despite topological perturbations, thereby ensuring more stable performance in disturbed environments. Enhancing these aspects will further elevate the algorithm's practical value in real-world scenarios, such as managing cross-platform public opinion to mitigate echo chambers or designing propagation-aware containment strategies for epidemic control.

7. Conclusion

In this paper, we propose a structure-aware cross-layer node representation algorithm to identify the optimal seed set for influence maximization in multilayer heterogeneous networks. A multilayer heterogeneous network independent cascade model is developed to characterize the intra-layer and inter-layer diffusion dynamics based on inter-layer connectivity density. Furthermore, a proxy reward function based on an expected influence estimator is designed. The proposed SACRL learns node representations through a graph transformer-based GNN encoder and employs a Dueling DQN to optimize the decision-making policy for seed selection. The performance of the proposed SACRL was evaluated on six real-world multilayer heterogeneous networks. Subsequent research will be dedicated to scaling the proposed SACRL to large-scale multilayer higher-order networks and applying the SACRL to address practical problems such as cross-platform public opinion control and epidemic spreading.

CRedit authorship contribution statement

Chenshuo Li: Writing – original draft, Validation, Software. **Jianxin Tang:** Writing – review & editing, Project administration, Methodology. **Lijun Liu:** Writing – review & editing, Visualization. **Ping Wang:** Supervision, Resources. **Binggang Ren:** Resources, Formal analysis. **Bingkun Liang:** Visualization, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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Author biography



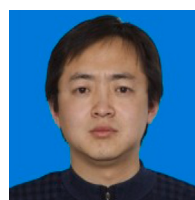
Jianxin Tang received the Ph.D. degree from the School of Information Science and Engineering, Lanzhou University, Lanzhou, China, in 2019. He is currently an Associate Professor with the School of Computer Science and Artificial Intelligence, Lanzhou University of Technology, Lanzhou, China. His work has focused on influence maximization in complex networks. His research interests include social computing, machine learning on graphs and social network analysis.



Chenshuo Li received the bachelor's degree in 2021, from the Tianjin Normal University, Tianjin, China. He is currently working toward the M.S. degree with the School of Computer Science and Artificial Intelligence, Lanzhou University of Technology, Lanzhou, China. His research interests include multilayer networks and reinforcement learning for influence maximization problem.



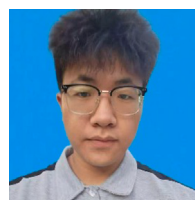
Lijun Liu received the bachelor's degree from Zhixing College of Northwest Normal University, China. She is currently pursuing the M.S. degree with the School of Computer Science and Artificial Intelligence, Lanzhou University of Technology, Lanzhou, China. Her research interests include intelligent optimization, multilayer social network analysis, and social computing.



Ping Wang received the Ph.D. degree from the School of Information Science and Engineering, Lanzhou University, Lanzhou, China, in 2024. He is currently an Associate Professor with the School of Traffic and Transportation, Lanzhou Jiaotong University, Lanzhou. His work has focused on influence maximization in social networks and transportation networks. His research interests include social network analysis and intelligent transportation systems.



Binggang Ren is a master's student at Lanzhou University of Technology. He received his B.S. degree from the School of New Engineering Industry at Putian University in 2025. His primary research focuses on the influence maximization problem in multilayer networks and the automatic design of metaheuristic algorithms.



Bingkun Liang is a master's student at Lanzhou University of Technology. He graduated in 2025 from the School of Computer Science at Weinan Normal University with a bachelor's degree. His primary research focuses on the metaheuristic algorithm and automatic algorithm design.